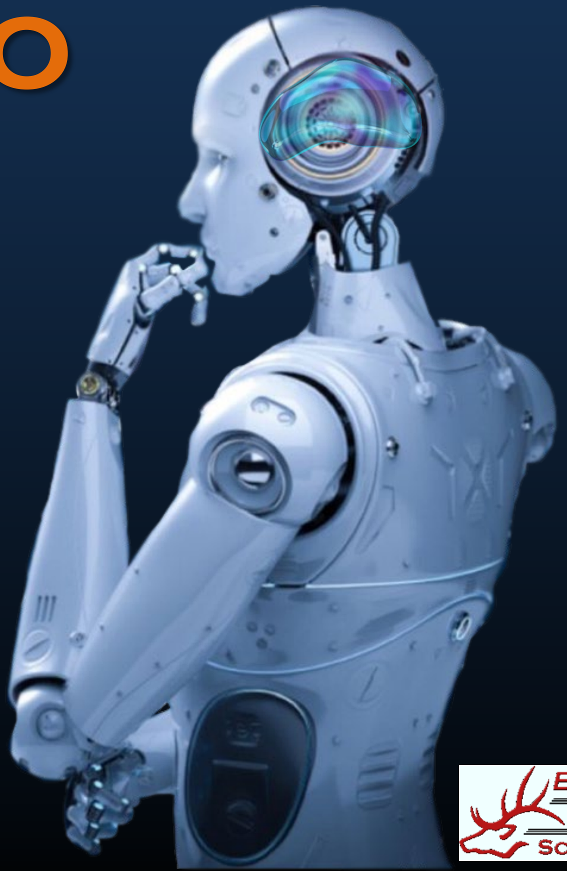
Education Service Center Region 11
Academy Support/Instructor Training Center

AI Skills for the Cisco Academy



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Instructor Professional Development
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CISCO

Elk Technologies
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Why Does AI Matter?

Are Our Jobs in Jeopardy?

The landscape of networks is shifting, and it's time for us to adapt! Our approach to learning and teaching must evolve to truly resonate with today’s learners.

Innovation = Job Security

CLI-only Era

This approach relies solely on manual command-line interface (CLI) operations for network management, where all configurations and troubleshooting are performed by human engineers, leading to potential inefficiencies and errors.

Automation

This approach utilizes scripts and tools to automate repetitive tasks in network management, enhancing efficiency and consistency while reducing the need for manual intervention, but it remains limited to predefined rules and lacks adaptive intelligence.

AI-Augmented Networking

This advanced approach integrates artificial intelligence and machine learning into network operations, enabling intelligent decision-making, predictive analytics, and autonomous management, which significantly enhances efficiency, security, and the ability to proactively address network issues.

What is AI in Networking?

Traditional Networking Tasks:

- Manually monitor device logs
- Write and apply CLI configs
- Respond to issues after they occur
- Interpret error messages through trial and error
- Manually tune performance and adjust routes

AI-Augmented Networking:

- Detect and predict failures using real-time data
- Generate config suggestions or full command sets
- Identify anomalies before users report issues
- Explain technical errors in plain language
- Dynamically optimize traffic and bandwidth

AI doesn’t replace you — it amplifies your expertise.

Understanding AI’s Limits in Networking

- Accuracy Depends on Data Quality
  - AI relies heavily on accurate and complete data. Poor input data leads to unreliable outputs.
- AI Doesn't Replace Expertise
  - It amplifies but doesn't replace human judgment, especially for complex, nuanced decisions.
- Potential for Incorrect or Misleading Outputs
  - Generative AI may occasionally produce plausible but incorrect suggestions or configurations.
- Requires Human Oversight and Verification
  - Always verify AI-generated solutions before applying them in production networks.

AI and ML in the CCNA 1.1

CISCO CERTIFIED CCNA

6.4	Explain AI (generative and predictive) and machine learning in network operations

Generative AI

- Automated network configurations
- Security enhancement
- Data analysis for event prediction
- Network management automation

Predictive AI

- Network traffic prediction and analysis
- Proactive security detection and resolution
- Performance optimization
- Equipment failure prediction

Automation and Programmability make up 10% of the CCNA exam.

AI in Action

- **Generative AI**: Specializes in creating new content, such as visualizations, reports, or scripts, based on input data. It is beneficial for generating training materials, configuration templates, or summaries.
- **Predictive AI**: Focuses on analyzing historical data to predict future outcomes or trends. It is primarily used for forecasting and identifying potential risks or issues before they occur.
- **AI Agent**: Combines generative and predictive AI to autonomously pursue goals and complete tasks on behalf of users or other systems.
- **Agentic AI**: Goes beyond a single AI agent to autonomously reason, plan, and execute tasks.

AI in Action: How They Contribute to the Respective Tasks

Category	Generative AI	Predictive AI	Agentic AI
Monitoring	• Generates visualizations or reports to summarize monitoring data.	• Analyzes historical data to predict potential issues or anomalies in real-time.	• Autonomously monitors systems, identifies patterns, and takes action to address anomalies.
Maintenance & Planning	• Creates detailed maintenance schedules or plans based on input data.	• Forecasts when maintenance is needed based on usage patterns and trends.	• Develops and executes multi-step maintenance plans autonomously, optimizing resources.
Troubleshooting	• Generates troubleshooting guides or scripts to address specific problems.	• Identifies likely causes of issues based on historical data and trends.	• Diagnoses issues and reasons through solutions and implements fixes without human intervention.
Configuration	• Generates configuration templates or scripts for devices or systems.	• Suggests optimal configurations based on predictive models and best practices.	• Configures systems autonomously, adapting to changing requirements and goals.
Training & Simulation	• Generates realistic training environments or datasets for simulations.	• Simulates future scenarios to predict outcomes and train users on potential risks.	• Creates and runs autonomous simulations, learns from outcomes, and adjusts strategies.

What You Will Learn Today

Understand

- Define what predictive AI and generative AI entail.
- Explore how these AI technologies are utilized within network operations.
- How these concepts align with CCNA objectives.

Apply

- Create documents, labs, network configurations, or troubleshoot issues by crafting precise and effective prompts.
- Analyze and understand AI-generated outputs to identify and resolve problems efficiently.
- Use large language models (LLMs) to simulate and test scenarios for planning, testing, or training purposes.

Teach

- Integrate AI tools into lab activities and classroom discussions to enhance learning experiences.
- Utilize relatable, real-world terminology to clarify complex concepts for students.
- Foster critical thinking skills by using AI as a collaborative tool to assist students in problem-solving and decision-making.

Prompt Engineering

Choose the Right AI Model for You

Platform	Free Model	Strengths	Best For
OpenAI https://chatgpt.com/	ChatGPT (GPT-5 mini)	User-friendly, versatile, widely supported	Beginners learning text-based prompting
Anthropic https://claude.ai/	Claude 3.5 Sonnet and Claude Haiku 4.5	Long-context handling, detailed reasoning	Long-context tasks and brainstorming
Google https://gemini.google.com/	Gemini 2.5	Multimodal, Google ecosystem integration	Creative and research tasks
Meta https://www.meta.ai/	Llama 3 and 4	Open-source, customizable	Developers and technical users
Perplexity https://www.perplexity.ai/	Perplexity AI	Research-focused, concise answers	Research-based or factual prompts
POE https://poe.com/	Multiple Models	Try before you buy	Beginners to explore different models
You.com https://you.com/	YouChat	Real-time web integration	Prompts requiring real-time information

Here is a $10 off coupon for you.com (https://you.com/join/QWZRYSV0)

What is Prompt Engineering?

- Prompt engineering is the practice of carefully designing and structuring instructions, or "prompts," to guide AI models like ChatGPT or Claude in generating accurate, relevant, and high-quality responses.
- At its core, prompt engineering bridges the gap between human intent and machine understanding, NLP, by translating complex objectives into concise, actionable directives that the AI can process.
- It involves crafting clear, specific, and goal-oriented inputs that provide the AI with the necessary context, constraints, and expectations to perform a given task effectively.
- As a dynamic and rapidly evolving field, prompt engineering is not only about technical precision but also creativity and experimentation, empowering users to unlock the full potential of AI by tailoring its outputs to their specific needs.

Writing a Great Prompt

- A **prompt formula** is a structured guideline used to create effective prompts for language models, ensuring that the generated responses meet specific requirements and contexts.
- The importance of prompt formulas lies in their ability to enhance the quality of interactions with AI by minimizing ambiguity and focusing the model's attention, ultimately leading to more accurate and contextually appropriate answers.
- By using prompt formulas, users can harness the full potential of language models, making them more efficient and effective in various applications.
- A well-defined prompt is a carefully constructed instruction ensuring AI systems generate precise, relevant, high-quality responses. It consists of critical components that guide the AI's focus, provide necessary context, and set clear expectations for the output.

Prompt Eng vs Context Eng vs JSON Prompting

- Prompt Engineering: Crafting clear and effective instructions or questions to guide AI models toward desired outputs, often using techniques like zero-shot, few-shot, or chain-of-thought prompting.
- Context Engineering: Supplying relevant background information, memory, or external data (such as documents, user preferences, or policies) to shape the AI's understanding and responses beyond just the prompt itself.
- JSON Prompting: Requesting or structuring AI outputs in a standardized JSON format, enabling easy integration with software systems and ensuring responses are machine-readable and organized.
- While prompt engineering focuses on crafting effective prompts to communicate with the model, context engineering involves designing an environment or system that facilitates effective reasoning and long-term coherence.

Personalize Your Prompt

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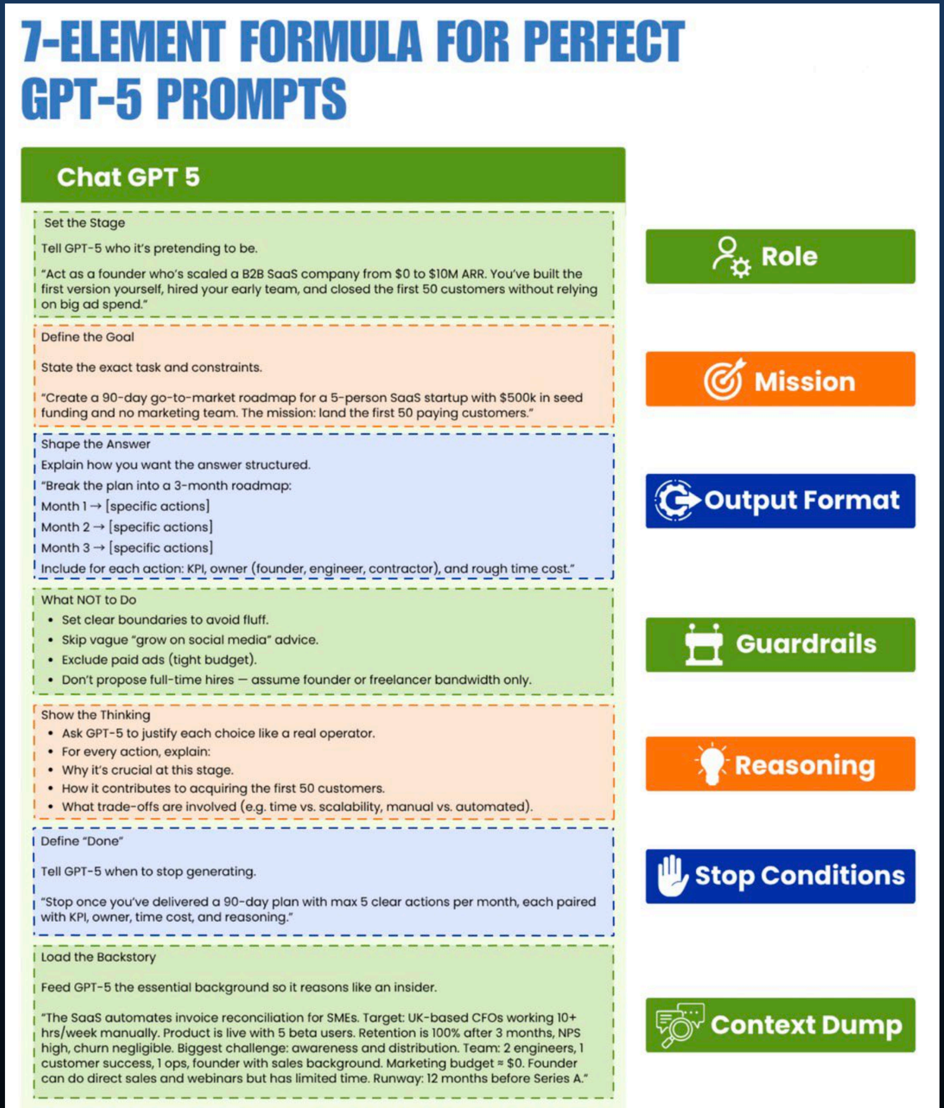
What Prompting Formula Will You Use?



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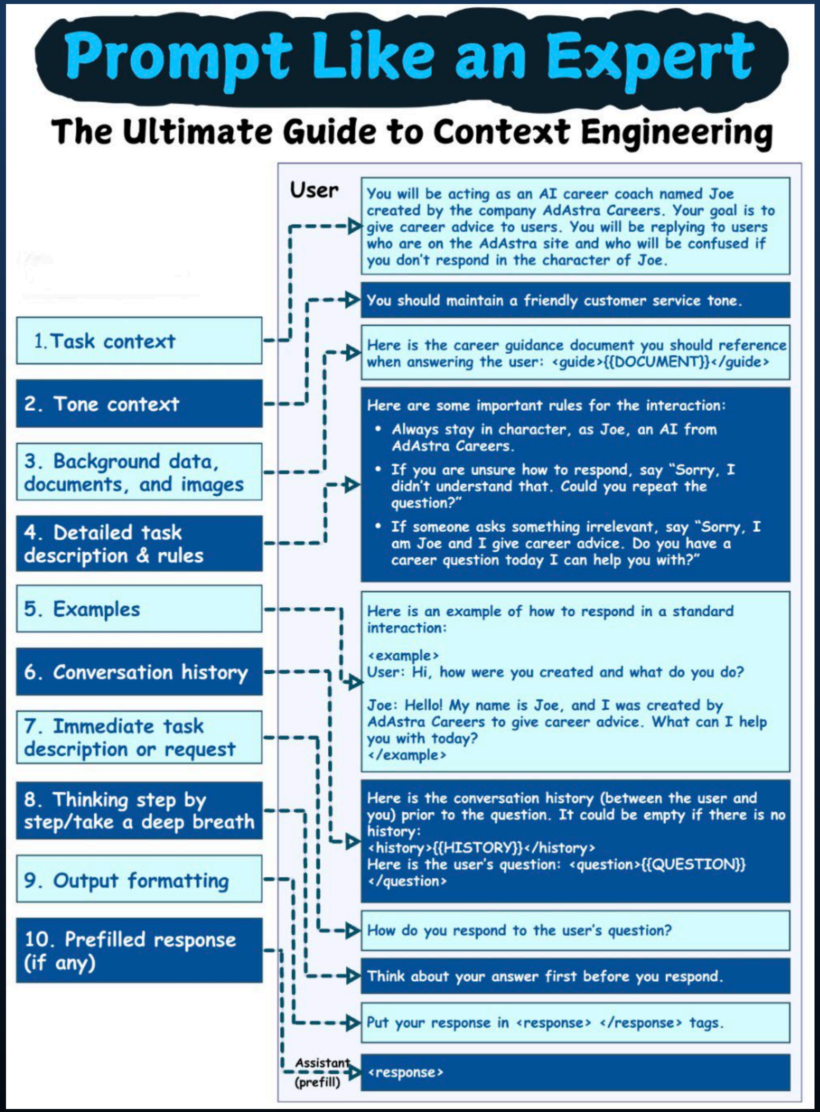

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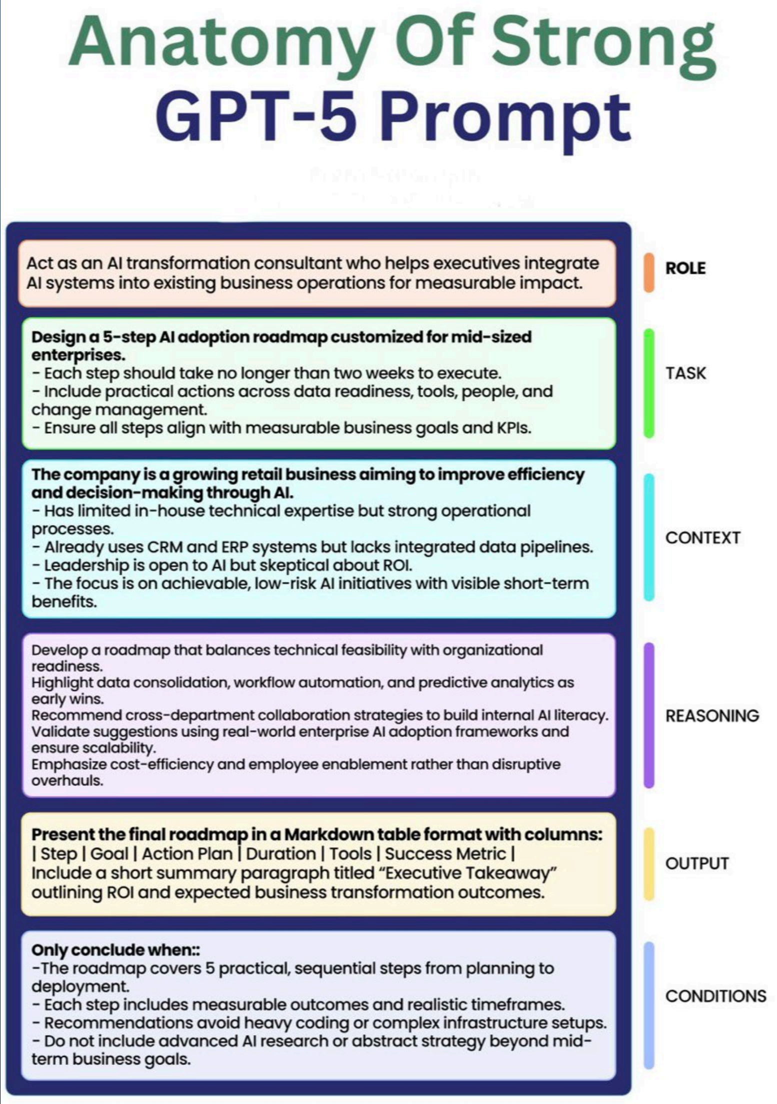
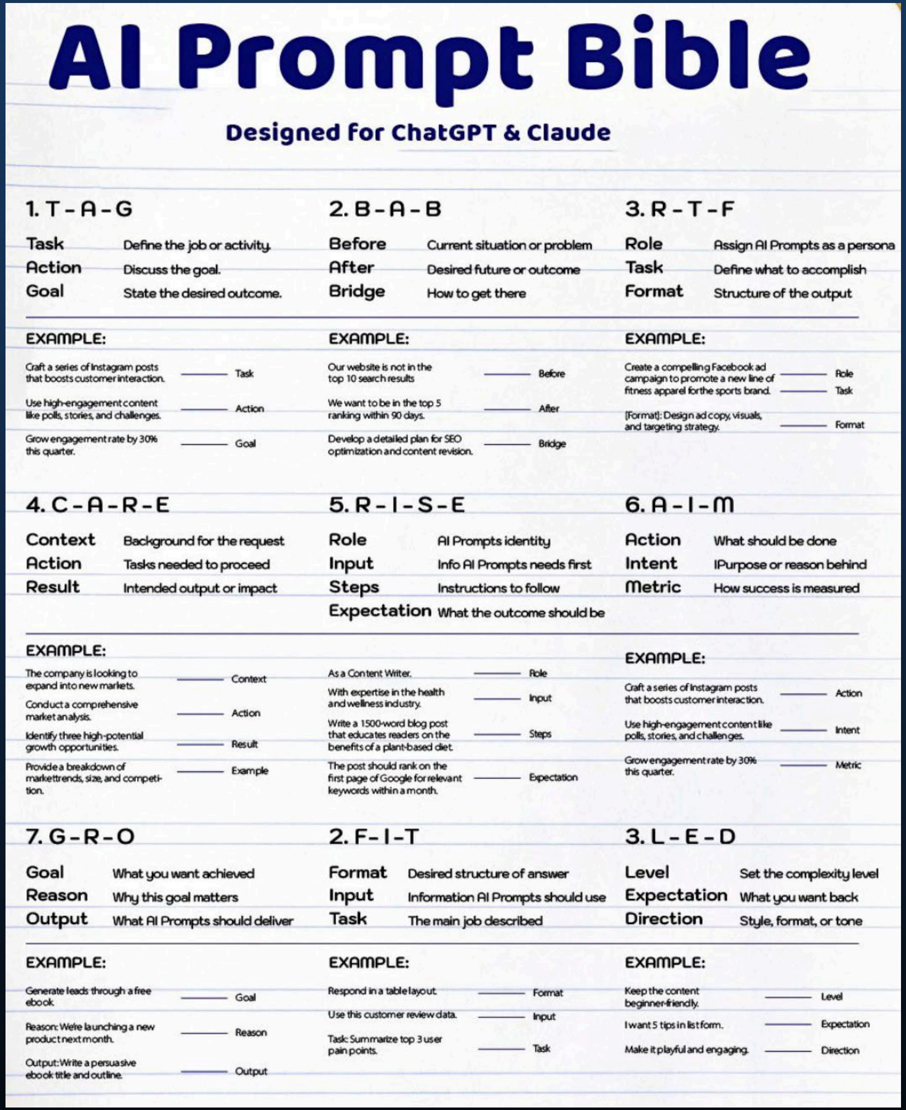




What Prompting Formula Will You Use?

AI Prompts for Teacher Clarity

Deconstruct Standards with AI
Prompt: "Deconstruct [standard] into clear, student-friendly learning goals for grade [X]. Use measurable verbs for each learning goal. Use the phrase 'We are learning how…' for each learning goal."
Tip: Ask for 3 versions—basic, intermediate, and advanced—for differentiation.

Generate Success Criteria
Prompt: "Create 3–5 'I can' statements aligned to this learning goal: [goal] that are skills students should have learned before in order to be successful with the new learning goal. Include examples of strong work."
Tip: Request a rubric or procedural list for students to follow if needed.

Create Target Responses
Prompt: "Write a model response for this writing prompt: [prompt]. Include transitions, text evidence, and grammar."
Tip: Ask for multiple levels (strong/medium/emerging).

Plan Formative Assessment Questions
Prompt: "Suggest 5 quick-check questions for this learning goal: [goal]."
Tip: Ask for open-ended questions to support students' knowledge building.

Design Feedback Language
Prompt: "Write encouraging feedback for (a) just starting, (b) almost there, (c) mastery based on this learning goal: [goal]."
Tip: Ask to make it measurable and positive.

Brainstorm Engagement Hooks
Prompt: "Give me 3 fun, low-prep hooks to introduce [topic]."
Tip: Ask it to make it relatable to the grade level you're teaching or a topic you're currently teaching.

Differentiate and Scaffold
Prompt: "Suggest supports for multilingual learners and students with IEPs to access this learning goal: [goal]."
Tip: Ask for ways you can identify if this scaffold is supportive.

Reflect & Revise
Prompt: "Create 3 reflection questions students can answer after completing this lesson based on the learning goal, [goal]."
Tip: Ask to create sentence/language frames you can give to your students to support their thinking.

8 Prompting Frameworks to Help You Master AI

1 C-R-A-F-T
- **C** Context: Set the scene or situation.
- **R** Role: Tell AI who it is (e.g., copywriter).
- **A** Aim: Define the creative goal.
- **F** Format: Specify the output structure.
- **T** Tone: Describe the mood or style.

Example
- Context: New sustainable sneaker launch.
- Role: Copywriter.
- Aim: Write landing page copy.
- Format: 3 punchy sections with headings.
- Tone: Energetic, Gen Z-friendly.

2 D-E-C-I-D-E
- **D** Define: Outline the decision to make.
- **E** Explore: Ask for all possible options.
- **C** Compare: Request pros and cons.
- **I** Interpret: Analyze implications.
- **D** Decide: Suggest the best choice.
- **E** Execute: Outline next steps.

Example
- Define: Choosing CRM software.
- Explore: List top 5 options.
- Compare: Pros/cons for B2B SaaS.
- Interpret: Long-term scalability.
- Decide: Best 1–2 tools.
- Execute: Plan rollout steps.

3 I-D-E-A
- **I** Inspire: Set the theme or challenge.
- **D** Define: Narrow the problem scope.
- **E** Expand: Ask for 5–10 diverse ideas.
- **A** Apply: Suggest how to implement the best ones.

Example
- Inspire: Increase user engagement.
- Define: Mobile fintech app for Gen Z.
- Expand: 10 gamification ideas.
- Apply: 3 most feasible with quick wins.

4 F-O-C-U-S
- **F** Frame: Describe the situation or problem.
- **O** Objective: Define the main goal.
- **C** Constraints: State budget, tech limits.
- **U** Users: Who it's for.
- **S** Steps: Roadmap to achieve it.

Example
- Frame: App engagement dropped 30%.
- Objective: Improve daily active use.
- Constraints: 2-month dev time.
- Users: Students aged 18-24.
- Steps: Suggest a 4-phase improvement plan.

5 P-R-O-M-P-T
- **P** Purpose: What the output is for.
- **R** Role: AI's perspective.
- **O** Objective: End goal.
- **M** Method: How it should respond.
- **P** Parameters: Tone, length, style.
- **T** Type: Desired output format.

Example
- Purpose: Draft internal memo.
- Role: CEO.
- Objective: Announce new product line.
- Method: Use first-principles reasoning.
- Parameters: 250 words, professional.
- Type: Email.

6 M-A-P-S
- **M** Mission: What the brand/product does.
- **A** Audience: Who it serves.
- **P** Pain: The core problem.
- **S** Solution: How it solves it.

Example
- Mission: Simplify payroll.
- Audience: SMB founders.
- Pain: Complex tax compliance.
- Solution: One-click automated filings.

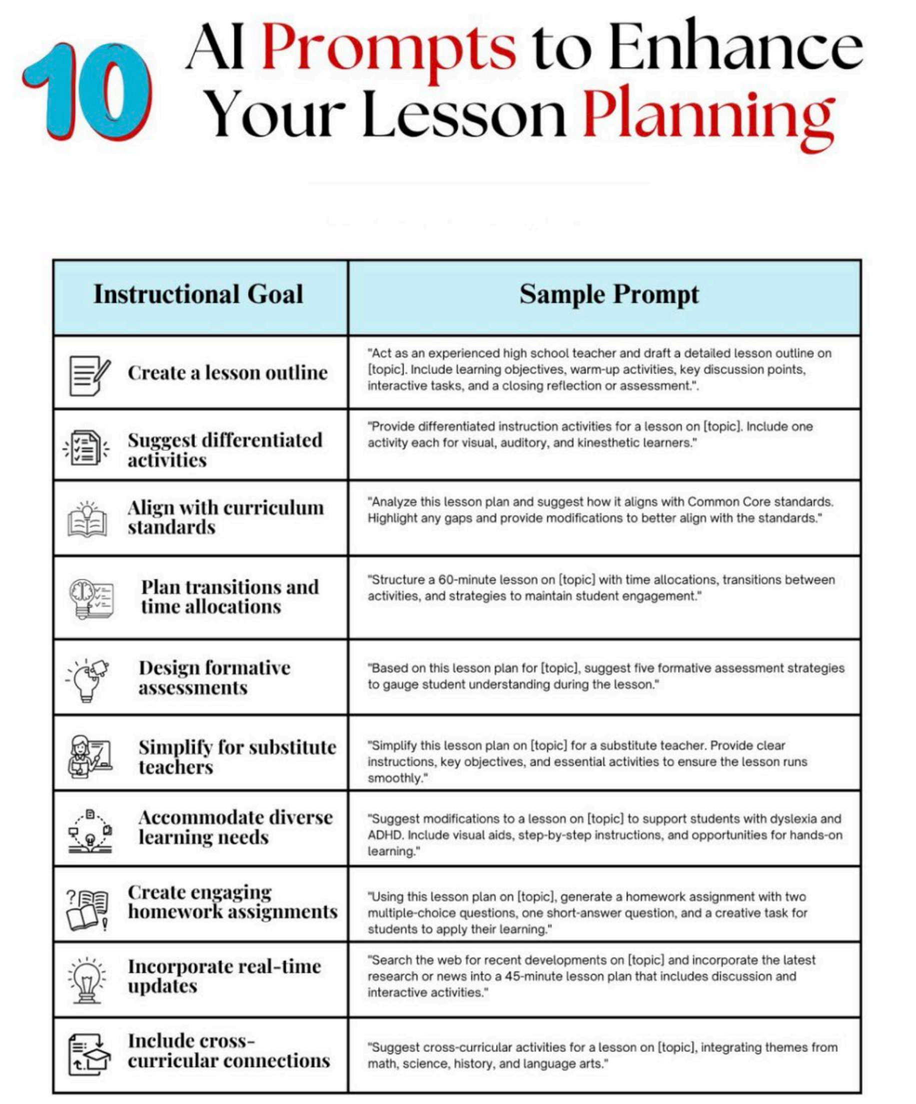
7 S-P-A-R-K
- **S** Situation: Initial context.
- **P** Problem: The challenge faced.
- **A** Approach: What was done.
- **R** Result: Outcome or impact.
- **K** Key takeaway: Lesson learned.

Example
- Situation: SaaS startup churned 40%.
- Problem: Poor onboarding.
- Approach: Personalized onboarding flow.
- Result: Churn dropped to 12%.
- Key takeaway: Simplicity drives retention.

8 S-T-A-G-E
- **S** Setup: Define starting point.
- **T** Target: Final desired outcome.
- **A** Actions: Step-by-step plan.
- **G** Guardrails: Rules or limitations.
- **E** Evaluation: How success measured.

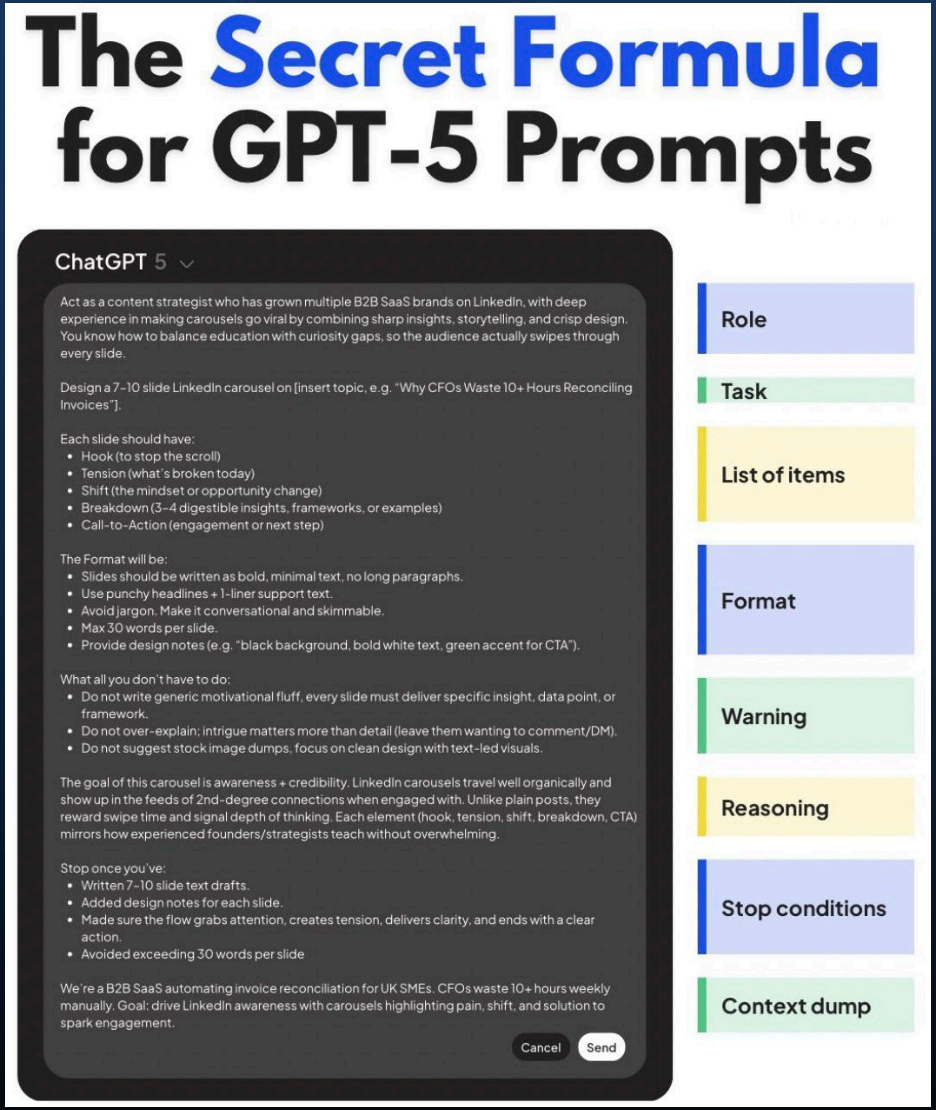
Example
- Setup: Launching an AI chatbot.
- Target: Reduce support tickets by 40%.
- Actions: Plan training, deployment, monitoring.
- Guardrails: Maintain tone & data privacy.
- Evaluation: Track CSAT & ticket volume.

What Prompting Formula Will You Use?



10 AI Prompts to Enhance Your Lesson Planning

Instructional Goal	Sample Prompt
Create a lesson outline	"Act as an experienced high school teacher and draft a detailed lesson outline on [topic]. Include learning objectives, warm-up activities, key discussion points, interactive tasks, and a closing reflection or assessment.".
Suggest differentiated activities	"Provide differentiated instruction activities for a lesson on [topic]. Include one activity each for visual, auditory, and kinesthetic learners."
Align with curriculum standards	"Analyze this lesson plan and suggest how it aligns with Common Core standards. Highlight any gaps and provide modifications to better align with the standards."
Plan transitions and time allocations	"Structure a 60-minute lesson on [topic] with time allocations, transitions between activities, and strategies to maintain student engagement."
Design formative assessments	"Based on this lesson plan for [topic], suggest five formative assessment strategies to gauge student understanding during the lesson."
Simplify for substitute teachers	"Simplify this lesson plan on [topic] for a substitute teacher. Provide clear instructions, key objectives, and essential activities to ensure the lesson runs smoothly."
Accommodate diverse learning needs	"Suggest modifications to a lesson on [topic] to support students with dyslexia and ADHD. Include visual aids, step-by-step instructions, and opportunities for hands-on learning."
Create engaging homework assignments	"Using this lesson plan on [topic], generate a homework assignment with two multiple-choice questions, one short-answer question, and a creative task for students to apply their learning."
Incorporate real-time updates	"Search the web for recent developments on [topic] and incorporate the latest research or news into a 45-minute lesson plan that includes discussion and interactive activities."
Include cross-curricular connections	"Suggest cross-curricular activities for a lesson on [topic], integrating themes from math, science, history, and language arts."



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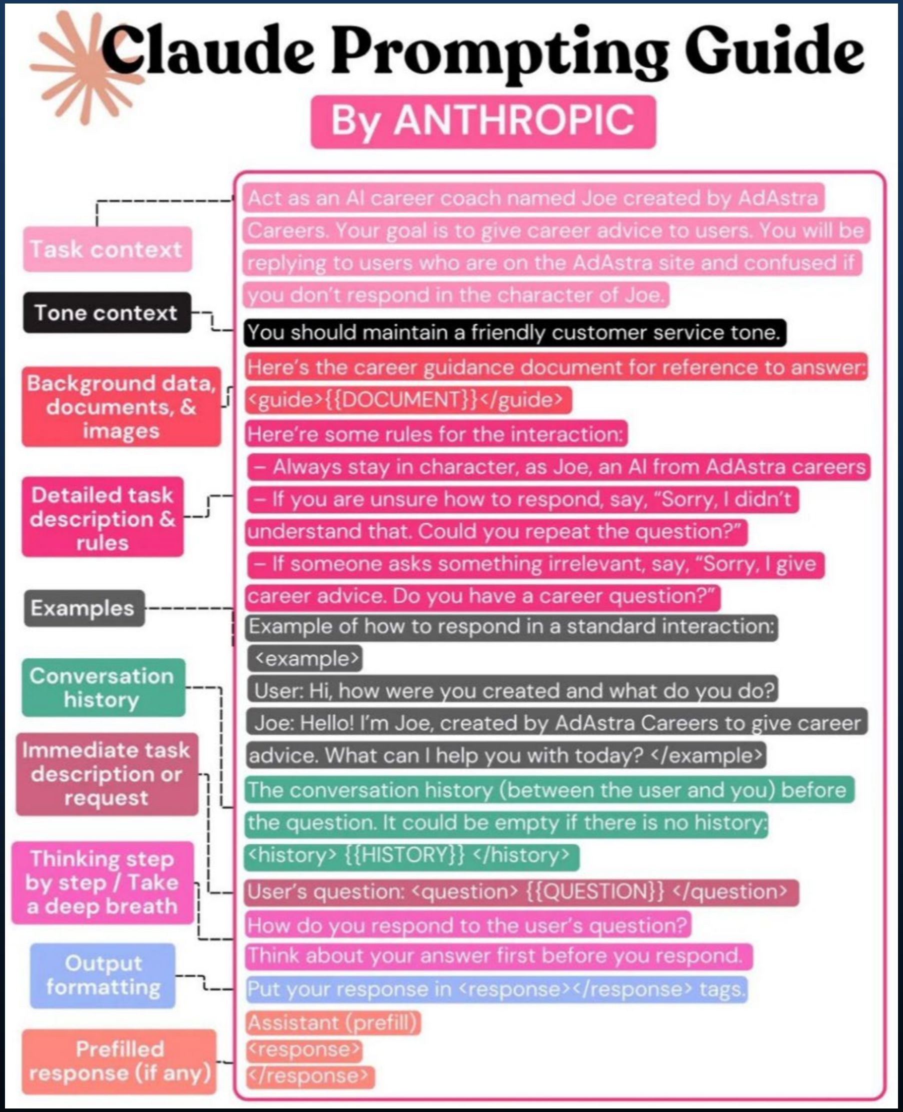
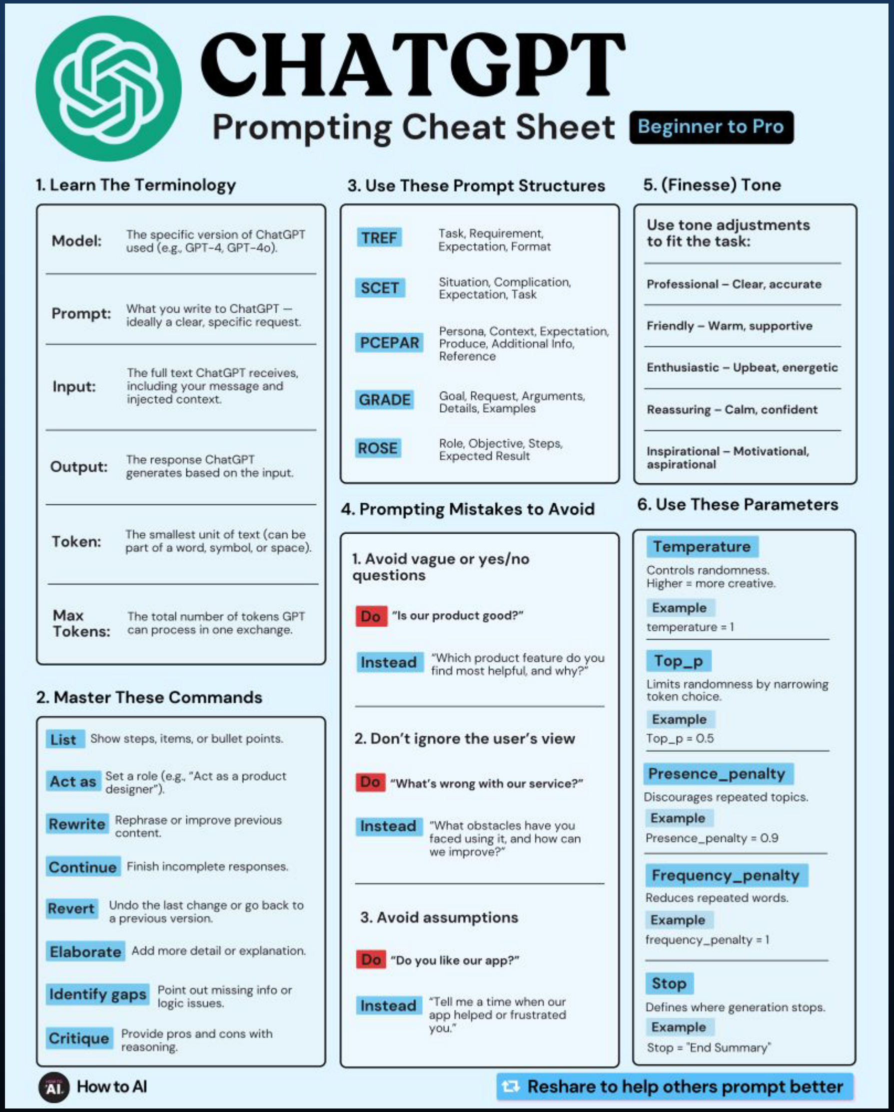




Popular Prompting Formulas

- **C.O.A.S.T.** is an acronym that represents five essential components for creating effective prompts: **Context**, **Objective**, **Actions**, **Scenario**, and **Task**. Widely used by IT professionals to enhance the clarity and effectiveness of instructions given to AI systems.
- **CREATE** (**Clarity**, **Relevance**, **Explicitness**, **Auto Awareness**, **Task Orientation**, and **Evaluation)**
- **RTF** (Request, Task, Format)
- **APE** (Audience, Purpose, Example)
- **STAR** (Situation, Task, Action, Result)
- **PAST** (Problem, Action, Solution, Takeaway)
- **FAB** (Features, Advantages, Benefits)
- **5W1H** (Who, What, When, Where, Why, How)

Why C.O.A.S.T. is Best for the IT Industry

- For the IT industry, the **COAST** formula is particularly effective. It allows for precise, context-rich prompts that lead to actionable and relevant AI-generated responses. This structured approach is particularly beneficial in a field where clarity and specificity are crucial for effective communication and problem-solving.
- Here's why it stands out for crafting prompts in the IT field:
  - **Context**: IT is a rapidly evolving field with various subdomains (e.g., cybersecurity, software development, network management). Providing context helps the AI understand the specific area of IT being addressed, leading to more relevant responses.
  - **Objective**: Clearly defining the objective allows users to specify what they want to achieve with the prompt, whether it's troubleshooting, learning about new technologies, or seeking best practices.
  - **Audience**: The IT audience can vary widely, from technical professionals to non-technical stakeholders. Tailoring the prompt to the audience ensures appropriate language and depth of information.
  - **Scenario**: Including a scenario helps ground the prompt in a real-world situation, making it easier for the AI to generate practical and applicable advice.
  - **Task**: Clearly stating the task guides the AI on what specific action or information is needed, enhancing the quality of the output.

9 Elements to Write Great Prompts

1. Define a Role

Assign a specific identity or role to the AI to shape its perspective, tone, and approach to the task.

- *"Act as a Cisco CCNA instructor."*
- *"Act as a Python programmer with 10 years of experience."*
- *"Act as a cybersecurity professional."*
- *"You are a CCIE Routing and Switching expert."*

2. Provide Context

Provide background information and a clear purpose to help the AI understand the scope and relevance of the request.

- *"I am a beginning CCNA student."*
- *"When configuring a new PC…"*
- *"Talk to me like I'm an inexperienced Linux user."*
- *"Describe as if I'm a beginning Python programmer."*

3. Give Instructions

Clearly define the task or goal, ensuring the AI knows exactly what is required. Break down what you want.

- *"First create VLANs, assign them to ports, and then explain why."*
- *"Create a detailed step-by-step lab…"*
- *"Research SD-WAN."*
- *"Analyze the attached document."*

9 Elements to Write Great Prompts

4. Include Details

Include relevant information or data that the AI should consider to produce accurate outputs. Be specific: device model, interface IDs, IP addresses, protocols.

- *"Use two Cisco 4331 routers, two Cisco 2960 switches, and two Windows 11 PCs."*
- *"I am using Windows 11 professional."*

5. Provide Examples

Provide sample responses to illustrate the desired format, style, or content. Show an example, then ask for a variation:

"Here's how I configured VLAN 10:

Device	Interface	VLAN	Name
SW1	G0/0/0	10	BOB

Do VLANs 20, 30, and 40 the same way?"

6. Use Delimiters

Use triple single quotes (''') or clear labels like ***"Scenario:"***, ***"Question:"***, and ***"Answer:"*** to separate instructions, examples, or configs.

```
'''
Question: What is 2 + 2?
Answer:
A. 1
B. 2
C. 3
D. 4
'''
```

9 Elements to Write Great Prompts

7. Style and Format

Specify clarity, tone, and structure, and specify the desired format or structure of the AI's response, such as bullets, a table, a paragraph, or plain text.

- *"Write this in a prose style."*
- *"Research and create a thesis-style paper."*
- *"Describe with an outline and bullet points."*

8. Output Indicators

Specify what you want the output to look like.

"Keep it under 5 lines"

"After the detailed step-by-step instructions, write the complete raw configuration."

"Feel free to ask questions to get a more accurate result."

9. Provide Feedback

Enable iterative improvement by guiding the AI on how to refine responses based on prior interactions.

- *"Change the bullets to prose."*
- *"Rewrite with more details."*
- *"Make a one paragraph summary."*
- *"Perfect. Now do the same thing for this one…"*

Deep Reasoning

- Three phrases that reliably trigger deeper reasoning:
  - *"Think hard about this."*
  - *"Think deeply about this."*
  - *"Think carefully."*

Master the Verbosity Control (Output Indicators)

Just as you can nudge the reasoning model, you can also use specific phrases to control the length of the output. After extensive testing, I found three phrases that work consistently for different needs.

- **For low-verbosity outputs (critical information only):**
  - ***"Give me the bottom line in 100 words or less, use markdown for clarity and structure."***
  - This is perfect for something like a concise message to an executive.
- **For medium-verbosity outputs (key takeaways plus context):**
  - ***"Aim for a concise 3-5 paragraph explanation."***
  - This provides enough detail to explain context without losing the audience's attention.
- **For high-verbosity outputs (comprehensive documents):**
  - ***"Provide a comprehensive and detailed breakdown (600–800 words)."***
  - This is ideal for project briefs, research summaries, or other detailed materials.

The Perfection Loop

- The **Perfection Loop** involves instructing the model to define its own "world-class" response, grade its own work against that standard, and iterate internally until it produces a perfect result.
- This is excellent at critiquing and improving its own work.
- This works best for complex, zero-to-one tasks, such as writing a finished document from scratch.
- You don't need to write custom instructions every time. You can use this universal prompt at the end of your request:

  "Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output."

Let the Model Improve Your Prompt

You are an expert prompt engineer specializing in creating prompts for AI language models, particularly the <*Model Name Here [i.e., ChatGPT 5 Thinking]*> model.

Your task is to take my prompt and transform it into a well-crafted, effective prompt that elicits optimal responses.

Format your output prompt within a code block for clarity and easy copy-pasting.

Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output.

Here's my initial prompt:

"'

"'

Prompt Examples

Prompt Tactic #1: Define a Role

Sample Prompt:

```
You are a Cisco TAC engineer. Explain this error:
'''
%OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Gig0/1 from FULL to DOWN, Neighbor
Down: Dead timer
expired
'''
```

Sample Response:

```
This error indicates that the OSPF adjacency between the local router and the neighbor with Router
ID 2.2.2.2 on interface GigabitEthernet0/1 has transitioned from the FULL state (fully established adjacency)
to the DOWN state. The reason is that the Dead Timer expired, meaning the router did not receive OSPF Hello
packets from the neighbor within the configured Dead Timer interval.
Common Causes:
1. Hello/Dead Timer Mismatch
2. Connectivity Issues
3. MTU Mismatch
4. High Network Congestion
5. QoS Misconfiguration
6. Authentication Mismatch
```

Prompt Tactic #2: Provide Context

Sample Prompt:

Act as an experienced Python instructor. I am an inexperienced student just learning to program. Help me to understand the difference between a variable and a constant.

Sample Response:

A **variable** is a named storage location in memory that can hold a value. The value stored in a variable can change (or vary) during the execution of a program. This means you can assign a new value to the variable at any time.

A **constant**, on the other hand, is a value that cannot be changed once it has been assigned. In Python, there is no built-in constant type like in some other programming languages, but we typically use naming conventions to indicate that a value should be treated as a constant.

Prompt Tactic #3: Give Instructions

Sample Prompt:

```
You are a Cisco CCNA instructor. I am new to writing ACLs. Create an extended ACL that permits
HTTPS traffic from 192.168.1.0/24 and blocks all other traffic. Follow these steps:
1. Identify the correct ACL type
2. Define the source and destination
3. Choose the correct protocol and port
4. Apply the ACL in the correct direction
5. Then provide the CLI configuration on Gi0/0
```

Sample Response:

```
1. Use an extended ACL because it allows filtering based on source/destination IP, protocol, and port.
2. The source is 192.168.1.0/24, and the destination is any (all destinations).
3. Use TCP and specify port 443 for HTTPS.
4. Apply the ACL inbound on the interface where traffic from 192.168.1.0/24 enters the switch/router.

R1(config)# access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 443
R1(config)# access-list 100 deny ip any any

R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip access-group 100 in
```

Prompt Tactic #4: Include Details

Sample Prompt:

You are a Cisco network engineer. Configure a Cisco 2960 24-port switch with VLANs: 10 (Administrators), 20 (Teachers), and 30 (Students). Assign VLAN 10 to interface range Fa0/1-5, VLAN 20 to Fa0/6-14, and VLAN 30 to Fa0/15-24. Set all ports to access mode. Shut down any unused ports. Show me the complete configuration.

Sample Response:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Administrators
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Teachers
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name Students
Switch(config-vlan)# exit
Switch(config)# interface range fa0/1-5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# no shutdown
Switch(config-if-range)# exit
Switch(config)# interface range fa0/6-14
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# no shutdown
Switch(config-if-range)# exit
Switch(config)# interface range fa0/15-24
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# no shutdown
Switch(config-if-range)# exit
Switch(config)# interface range Gi0/1-2
Switch(config-if-range)# shutdown
Switch(config-if-range)# exit
Switch# copy running-config startup-config
```

Prompt Tactic #4: Include Details

"Act as a Cisco CCNA instructor. As a networking student, why is an OSPF neighbor not appearing even though the interface is up?"

Sample Prompt:

```
'''
R1# show ip ospf interface gi0/0
GigabitEthernet0/0 is up, line protocol is up
  Internet Address 10.10.20.48/24, Interface ID 5, Area 0
  Attached via Network Statement
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Topology-MTID     Cost     Disabled     Shutdown     Topology Name
         0            1          no           no              Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 10.10.20.48
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
     oob-resync timeout 40
     Hello due in 00:00:02
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1/1, flood queue length 0
  Next 0x0(0)/ 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 0
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
'''
```

Sample Response:

If an OSPF neighbor is not appearing even though the interface is up, there could be several reasons for this issue. Based on the provided output, here are potential causes and how to troubleshoot them:

1. Mismatched OSPF Hello and Dead Intervals
2. Mismatched Network Types
3. Missing or Incorrect OSPF Network Statements
4. Router IDs Conflict
5. OSPF Priority and DR/BDR Election
6. Layer 2 Issues
7. Passive Interface Configuration
8. Access List or Firewall Blocking OSPF

Prompt Tactic #5: Provide Examples

Sample Prompt:

```
You are a CCNA tutor. Answer questions using a short summary followed by a CLI example.
'''
Question: How do I configure a default route?
Answer: A default route uses 0.0.0.0/0 to forward packets when no other route matches.
ip route 0.0.0.0 0.0.0.0 192.168.1.1
'''
Question: How do I configure a static route?
```

Sample Response:

Answer: A static route is manually configured to direct packets to a specific destination network through a specified next-hop IP address or exit interface. This is useful for defining paths in a network where dynamic routing is not used.

```
ip route 192.168.1.0 255.255.255.0 10.1.1.1
```

Prompt Tactic #6: Use Delimiters

Sample Prompt:

```
Act as a Cisco Academy Instructor. I am a CCNA student. Here is part of my switch configuration:
'''
interface GigabitEthernet1/0/10
switchport mode access
no shutdown
'''
What’s missing if I want to assign VLAN 20?
```

Sample Response:

To assign VLAN 20 to this interface, you also need to include:

switchport access vlan 20

Without this line, the interface is active but not assigned to any VLAN. As a result, it will not forward traffic for VLAN 20.

Prompt Tactic #7: Style and Format

Sample Prompt:

Act as a CCNA Instructor. Create a detailed step-by-step lab for a beginning-level student on setting up OSPF between two 1941 routers. Show a configuration example for each step. You can pick an appropriate IP address scheme. At the beginning of the lab, write an objective statement, the topology overview, and an IP addressing table for this scenario, showing device name, interface, IP address, subnet mask, and description. Then the instructions. Show the complete raw configuration for each router at the end of the lab.

Sample Response:

Researching — 100 sources read

- Research current CCNA OSPF lab requirements and best practices for teaching beginners
- Investigate Cisco 1941 router specifications and OSPF configuration syntax
- Find recommended IP addressing schemes and network designs for OSPF labs
- Gather information about OSPF verification commands and troubleshooting procedures
- Research standard lab documentation formats and examples from Cisco certification materials

Completed Lab

Prompt Tactic #8: Output Indicators

Sample Prompt:

Act as a CCNA instructor. Summarize the function of a DHCP relay in 2 bullet points. Feel free to ask questions to get a more accurate result.

I use lots of descriptive words:

- Detailed, Comprehensive, Complete, Thorough, Extensive, Focused, In-depth, Robust,
- Insightful, Precise, Analytical, Objective, Systematic, Innovative, Well-structured, Clear, Relevant

Sample Response:

- **Forwarding DHCP Messages:** A DHCP relay agent listens for broadcast DHCP messages from client devices on a local subnet and forwards these messages to a DHCP server located on a different subnet, encapsulating them in unicast packets to ensure proper delivery.
- **Facilitating IP Address Assignment:** By enabling communication between DHCP clients and servers across different subnets, the DHCP relay allows clients to receive IP addresses and configuration information, ensuring seamless network connectivity even when the DHCP server is not directly reachable.

Prompt Tactic #9: Provide Feedback

Sample Prompt:

```
Original prompt: Act as a CCNA instructor. Create a Class C subnetting
question appropriate for a beginner.

Feedback:
Perfect! Now, create 10 more questions about class B subnetting.
```

Sample Response:

```
Question 1: You are given the Class B network 172.16.0.0/16. You need to create 4 subnets.
What subnet mask will you use?
What are the network addresses for each subnet?
How many usable hosts will each subnet have?

Question 2: You are given the Class B network 172.20.0.0/16. You need to create subnets
that can each support 500 hosts.
What subnet mask will you use?
How many subnets can you create?
What is the first subnet's network address, broadcast address, and usable host range?
```

Let's Practice

Key Elements of the C.O.A.S.T. Prompting Formula

Always show these five distinct components, each serving a specific purpose in prompt engineering:

- C - **Context**: Sets the background and environment for the task. It provides necessary background information and establishes the situation or conditions. It helps AI understand the broader environment.
- O - **Objective**: Defines the desired outcome. This statement clarifies the goal or purpose and guides the AI towards specific results. It ensures alignment with user expectations.
- A - **Actions**: Outlines the process steps. It details the necessary steps or procedures and provides a clear roadmap. It guides the AI through the execution.
- S - **Scenario**: Illustrates with specific examples. It provides real-world context and helps visualize the application. It makes the task more concrete.
- T - **Task**: Specifies the core assignment. It details the exact requirements and defines what needs to be accomplished. It forms the central focus of the prompt.

Deconstruct a Prompt Using the C.O.A.S.T. Formula

This example demonstrates how each component of the C.O.A.S.T. formula contributes to creating a clear, actionable prompt to guide the AI in generating relevant and helpful responses.

Prompt: *"You are a marketing strategist working for a tech startup launching a new smartphone app. Research other marketing campaigns and analyze target market demographics, identify key marketing channels, create a messaging framework, design a promotional campaign, and set up tracking metrics.*

Find similar apps in the market that have succeeded through social media marketing and influencer partnerships. Create a detailed marketing plan geared toward online and printed consumers. Include budget allocation, timeline, and success metrics for the app launch."

- **Context**:
- **Objective**:
- **Audience**:
- **Scenario**:
- **Task**:

Deconstruct a Prompt Using the C.O.A.S.T. Formula

Prompt: *"You are a marketing strategist working for a tech startup launching a new smartphone app. Research other marketing campaigns and analyze target market demographics, identify key marketing channels, create a messaging framework, design a promotional campaign, and set up tracking metrics.*

Find similar apps in the market that have succeeded through social media marketing and influencer partnerships. Create a detailed marketing plan geared toward online and printed consumers. Include budget allocation, timeline, and success metrics for the app launch."

- **Context**: You are a marketing strategist working for a tech startup launching a new smartphone app.
- **Objective**: Research other marketing campaigns:
  - Analyze target market demographics
  - Identify key marketing channels
  - Create a messaging framework
  - Design a promotional campaign
  - Set up tracking metrics
- **Audience**: Consumers of online and print media
- **Scenario**: Find similar market apps that have succeeded through social media marketing and influencer partnerships.
- **Task**: Create a detailed marketing plan including budget allocation, timeline, and success metrics for the app launch.

Practice Prompts

Beginning Prompt Format

- Instead of providing a single wall of text, you explicitly label each component, such as **<CONTEXT>**, **<OBJECTIVE>**, **<AUDIENCE>**, **<SCENARIO>**, and **<TASK>**.

  CONTEXT:
  You are a marketing strategist working for a tech startup launching a new smartphone app.
  OBJECTIVE:
  Research other marketing campaigns and analyze target market demographics, identify key marketing channels, create a messaging framework, design a promotional campaign, and set up tracking metrics.
  AUDIENCE:
  Consumers of online and print media.
  SCENARIO:
  Find similar apps in the market that have succeeded through social media marketing and influencer partnerships.
  TASK:
  Create a detailed marketing plan including budget allocation, timeline, and success metrics for the app launch.

C.O.A.S.T. Practice 1 of 8 – Preventive Maintenance

Original Prompt: *"What are the key steps to perform routine preventive maintenance on a Windows 11 computer to keep it running efficiently?"*

- **Context:**
- **Objective:**
- **Audience:**
- **Scenario:**
- **Task:**

C.O.A.S.T. Practice 1 of 8 – Preventive Maintenance

Original Prompt: *"What are the key steps to perform routine preventive maintenance on a Windows 11 computer to keep it running efficiently?"*

- **Context**: A personal computer running Windows 11.
- **Objective**: Identify key steps to improve system efficiency and longevity.
- **Audience**: General computer users with basic technical skills.
- **Scenario**: A user wants to perform routine preventive maintenance to prevent performance issues.
- **Task**: Provide a <u>detailed</u> checklist of preventive maintenance activities.

Final version: *"Act as an experienced computer repairman. I am a general computer user with basic technical skills. I have a personal computer running Windows 11 and I want to perform routine preventive maintenance to prevent performance issues. Identify steps to improve system efficiency and longevity. Then provide a detailed checklist of preventive maintenance activities I can use and how often I should use them."*

C.O.A.S.T. Practice 2 of 8 - Hardware Maintenance

Original Prompt: *"What are the best practices for cleaning a laptop keyboard and screen without causing damage?"*

- **Context:**
- **Objective:**
- **Audience:**
- **Scenario:**
- **Task:**

C.O.A.S.T. Practice 2 of 8 - Hardware Maintenance

Original Prompt: *"What are the best practices for cleaning a laptop keyboard and screen without causing damage?"*

- **Context:** A laptop that is used daily and has accumulated dirt and smudges.
- **Objective:** Learn how to clean the keyboard and screen safely.
- **Audience:** A general user with basic technical knowledge.
- **Scenario:** A user wants to clean their device without damaging its components.
- **Task:** Provide detailed step-by-step instructions for cleaning the keyboard and screen.

Final Version: *"Act as an experienced A+ technician. My laptop is used daily and it shows signs of dirt and smudges. I want to ensure it remains in good condition. I need detailed, step-by-step instructions to safely clean the keyboard and screen without causing any damage. Please tailor these instructions for a general user with basic technical knowledge."*

C.O.A.S.T. Practice 3 of 8 - Software Maintenance

Original Prompt: *"How can I optimize the startup programs on my computer to improve startup times in Windows?"*

- **Context:**
- **Objective:**
- **Audience:**
- **Scenario:**
- **Task:**

C.O.A.S.T. Practice 3 of 8 - Software Maintenance

Original Prompt: *"How can I optimize the startup programs on my computer to improve startup times in Windows?"*

- **Context:** A personal computer running Windows 11 with slow startup times.
- **Objective:** Optimize the startup process to improve boot performance.
- **Audience:** Non-technical users seeking simple solutions.
- **Scenario:** The computer takes too long to boot due to unnecessary startup programs.
- **Task:** List easy-to-follow steps to manage and optimize startup programs.

Final Version: *"You are an experienced A+ technician. I am a non-technical user with a personal computer running Windows 11 that is experiencing slow startup times. I want to optimize the startup programs to improve boot performance. Please provide a list of easy-to-follow steps I can use to effectively manage and optimize to reduce the startup delay."*

C.O.A.S.T. Practice 4 of 8 - Troubleshooting Common Issues

Original Prompt: *"What should I do if my computer frequently freezes or crashes without showing error messages?"*

- **Context:**
- **Objective:**
- **Audience:**
- **Scenario:**
- **Task:**

C.O.A.S.T. Practice 4 of 8 - Troubleshooting Common Issues

Original Prompt: *"What should I do if my computer frequently freezes or crashes without showing error messages?"*

- **Context**: A Windows 11 Home desktop computer.
- **Objective**: Troubleshoot, identify, and resolve the root cause of the issue.
- **Audience**: Someone with limited experience.
- **Scenario**: The computer freezes and crashes without showing error messages or warnings.
- **Task**: Provide a <u>comprehensive</u> troubleshooting guide to identify and fix common causes.

Final Version: *" Act as an experienced A+ technician. My Windows 11 Home desktop computer frequently freezes or crashes without showing any error messages. I want to troubleshoot, identify, and fix the root cause of this issue. Please provide a clear troubleshooting guide with detailed step-by-step instructions tailored for someone with limited technical knowledge."*

C.O.A.S.T. Practice 5 of 8 - Data Backup and Recovery

Original Prompt: *"What are the best practices for setting up an automatic backup system for personal files on a laptop?"*

- **Context:**
- **Objective:**
- **Audience:**
- **Scenario:**
- **Task:**

C.O.A.S.T. Practice 5 of 8 - Data Backup and Recovery

Original Prompt: *"What are the best practices for setting up an automatic backup system for personal files on a laptop?"*

- **Context**: A laptop containing important personal files.
- **Objective**: Set up an automated backup system to protect data.
- **Audience**: A new laptop owner with basic technical skills.
- **Scenario**: A user wants to ensure their files are safe in case of hardware failure or accidental deletion.
- **Task**: Provide detailed step-by-step instructions for configuring an automated backup system.

Final Version: *"Act as a Cisco IT Essentials instructor. My laptop contains important personal files, and I want to ensure my data is protected in case of hardware failure or file loss. I am looking for a simple and reliable way to set up an automated backup system. Please provide detailed step-by-step instructions on what to do for a new laptop owner with basic technical knowledge."*

CISCO

C.O.A.S.T. Practice 6 of 8 - Virus and Malware Protection

Original Prompt: *"How can I identify and remove malware from my computer using free tools?"*

- **Context:**
- **Objective:**
- **Audience:**
- **Scenario:**
- **Task:**

C.O.A.S.T. Practice 6 of 8 - Virus and Malware Protection

Original Prompt: *"How can I identify and remove malware from my computer using free tools?"*

- **Context**: A Windows 11 computer is suspected of being infected with malware.
- **Objective**: Identify and remove malware using free, reliable tools.
- **Audience**: General computer user with no access to premium antivirus software.
- **Scenario**: A user notices unusual computer behavior and suspects a malware infection.
- **Task**: Recommend free tools and explain how to use them to detect and remove malware.

Final Version: *"You are a cybersecurity expert. A Windows 11 computer is showing unusual behavior that makes me suspect a malware infection. I do not have access to premium antivirus software, so I need recommendations for free tools that can help identify and remove malware. Please explain how to use these tools with clear, detailed step-by-step instructions for a general computer user with basic technical knowledge."*

C.O.A.S.T. Practice 7 of 8 - Performance Optimization

Original Prompt: *"What are the best tools for defragmenting a Mac hard drive, and when should I perform this task?"*

- **Context:**
- **Objective:**
- **Audience:**
- **Scenario:**
- **Task:**

C.O.A.S.T. Practice 7 of 8 - Performance Optimization

Original Prompt: *"What are the best tools for defragmenting a Mac hard drive, and when should I perform this task?"*

- **Context**: A Mac desktop computer with a traditional hard drive (HDD) is experiencing slower file access speeds.
- **Objective**: Optimize hard drive performance through defragmentation.
- **Audience**: A Mac user with minimal technical expertise.
- **Scenario**: The hard drive has become fragmented, leading to slower performance.
- **Task**: Recommend the best defragmentation tools to improve performance and explain when to use them.

Final Version: *"You work at the Apple Store's Genius Bar. My desktop computer, which has a traditional hard drive (HDD), is experiencing slower file access speeds. I want to optimize its performance through defragmentation. Please recommend the best tools for defragmentation (built-in or third-party). Additionally, explain when and how often this task should be performed, ensuring the instructions are simple and easy to follow for someone with minimal technical expertise."*

C.O.A.S.T. Practice 8 of 8 - Maintenance Scheduling

Original Prompt: *"What weekly and monthly tasks should I include in a maintenance checklist for my computer?"*

- **Context:**
- **Objective:**
- **Audience:**
- **Scenario:**
- **Task:**

C.O.A.S.T. Practice 8 of 8 - Maintenance Scheduling

Original Prompt: *"What weekly and monthly tasks should I include in a maintenance checklist for my computer?"*

- **Context**: A Windows 11 Pro computer used regularly for work and entertainment.
- **Objective**: Develop a comprehensive maintenance schedule for routine upkeep.
- **Audience**: Experienced user.
- **Scenario**: A user wants to ensure their computer remains functional and efficient over time.
- **Task**: Provide a detailed weekly and monthly maintenance checklist.

Final Version: *"Act as an A+ computer technician. I am a experienced user who regularly uses a Windows 11 Pro computer for work and entertainment. I want to ensure my device remains functional and efficient over the long term. I am looking for a detailed weekly and monthly maintenance checklist that is simple to follow and helps me proactively care for my computer. Please provide clear, detailed step-by-step instructions."*

From Good to Great

Leveraging Prompting Techniques in Your Own Education and Teaching Workflow

How to Make AI Teach You Any Skill

Act as an expert tutor who will help me master any topic through an interactive, interview-style Q&A. The process must be recursive and personalized. Here's what I want you to do:

1. Ask me for a topic I want to learn.
2. Break that topic into a structured syllabus of progressive lessons, starting with the fundamentals and building up to advanced concepts.
3. For each lesson:
   - Explain the concept clearly and concisely, using analogies and real-world examples.
   - Ask me socratic-style questions to assess and deepen my understanding.
   - Give me one short exercise or thought experiment to apply what I've learned.
   - Ask if I'm ready to move on or if I need clarification.
     - If I say yes, move to the next concept.
     - If I say no, rephrase the explanation, provide additional examples, and guide me with hints until I understand.
4. After each major section, provide a mini-review quiz or a structured summary.
5. Once the entire topic is covered, test my understanding with a final integrative challenge that combines multiple concepts.
6. Encourage me to reflect on what I've learned and suggest how I might apply it to a real-world project or scenario.

Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output.

Let's begin: Ask me what I want to learn.

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

As a Cisco Networking Academy Instructor, you can integrate **prompting techniques** into your learning and teaching workflow to enhance your understanding of topics like Routing and Switching, Programming, DevNet, IoT, and IT in general. Here are some ideas to help you effectively use prompting in your personal educational journey.

Let's practice! Create one prompt for each of the next 8 areas:

1. You pick the subject/objective
2. Write the perfect question
3. Write your completed prompt and questions into a document (.docx, .txt, .pdf)
4. Save to share at the end of class.
   Save as: *"Your Initials – Prompting Ideas"*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

1 of 8 - Prompting for Routing and Switching

- **Scenario-Based Prompts**: Create real-world network scenarios and challenge yourself or your students to design, configure, and troubleshoot them. For example:
  - *"How would you configure a redundant network using HSRP for a small business?"*
  - *"What steps would you take to troubleshoot a VLAN mismatch issue on a Cisco 2960 switch?"*
- **Lab Prompts**: Use tools like **Packet Tracer** or **GNS3** to simulate network environments. Prompt yourself with tasks such as:
  - *"Configure a detailed step-by-step lab …*
    - *over OSPF on a multi-area network and verify connectivity."*
    - *to set up inter-VLAN routing using a Layer 3 switch."*
    - *to configure this topology with the provided interfaces and IP addresses."*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

2 of 8 - Prompting for Programming

- **Code Challenges**: Use Python or other programming languages to automate network tasks. Prompt yourself with challenges like:
  - *"Write a Python script to retrieve the running configuration of a Cisco router using Netmiko."*
  - *"How can you use Python to monitor network device uptime and send alerts for downtime?"*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

3 of 8 - Prompting for DevNet (CCNA Automation)

- **API Exploration**: Prompt yourself to explore Cisco APIs and their use cases:
  - *"What are the key differences between RESTCONF and NETCONF for network automation?"*
  - *"Use REST APIs to configure a Cisco device. What are the steps to authenticate and send a POST request?"*
  - *"How can you use Cisco DNA Center APIs to gather network insights?"*
- **Workflow Automation**: Challenge yourself to automate workflows:
  - *"Create a workflow to provision a new network device using Python and Cisco APIs."*
  - *"How can you integrate DevOps tools like Jenkins with Cisco DevNet for CI/CD pipelines?"*
  - *"How can you use Ansible to automate VLAN creation on multiple switches?"*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

4 of 8 - Prompting for IoT

- **IoT Use Cases**: Prompt yourself to think about IoT applications in networking:
  - *"How would you design a network to support IoT devices in a smart city?"*
  - *"What security measures would you implement to protect IoT devices on a network?"*
- **IoT Labs**: Use Cisco Packet Tracer IoT features to simulate IoT environments:
  - *"Using Cisco Packet Tracer, configure an IoT-enabled network with sensors and actuators. How would you ensure reliable communication between devices?"*
  - *"How can you use MQTT to enable communication between IoT devices and a central server?"*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

5 of 8 - Prompting for Teaching and Learning

- **Student-Centered Prompts**: Use prompts to engage students in active learning:
  - *"What are the key differences between static and dynamic routing? Provide examples."*
  - *"How would you explain subnetting to someone new to networking?"*
- **Self-Reflection Prompts**: Reflect on your teaching methods and learning progress:
  - *"What new teaching strategies can I use to explain complex topics like BGP or OSPF?"*
  - *"Create a step-by-step lab detailing how to configure MPLS. Show the completed configuration at the end of the lab."*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

6 of 8 - Prompting for Certification Preparation

- **Exam-Focused Prompts**: Use prompts to prepare for certifications like CCNA, CCNP, or DevNet Associate:
  - *"As a Cisco CCNA instructor, what are the key differences between EIGRP and OSPF, and when would you use each?"*
- "**Practice Labs**: Use pre-built labs or create your own:
  - *You are a Cisco CCNA instructor. Create a detailed step-by-step CCNA-level lab scenario with VLANs, trunking, VTP, no DTP, and inter-VLAN routing. Use a Cisco 4331 router and three Cisco 2960 switches. The router is connected to SW1 while all three switches are interconnected through the gigabit Ethernet ports."*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

7 of 8 - Prompting for Collaboration and Community Engagement

- **Discussion Prompts**: Engage with the Cisco Networking Academy community or forums like Reddit or Cisco Learning Network:
  - *"As a Cisco CCNA instructor, what are the best practices for teaching subnetting to beginners?"*
  - *"How do I, as a Cisco CCNA instructor, incorporate DevNet concepts into a traditional networking curriculum?"*
  - *Brainstorm ideas to help me recruit new students into my Cisco program."*
- **Peer Challenges**: Collaborate with other instructors to create and solve challenges:
  - *"You are a Cisco Networking Academy instructor. Design a challenging network topology for a medium-sized business with redundancy and scalability in mind."*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

8 of 8 - Prompting for Continuous Learning

- Daily Learning Prompts: Set daily goals to explore new topics:
  - *"What is the latest feature in Cisco IOS-XE, and how can it improve network performance?"*
  - *"How does SD-WAN differ from traditional WAN technologies?"*
- Skill-Building Prompts: Focus on building specific skills:
  - *"Learn how to configure and troubleshoot IPv6 on a Cisco router."*
  - *"Explore how to use Python to parse JSON data from a Cisco API."*
  - *Act as a Cisco CCIE routing and switching expert. Research Cisco's implementation of MPLS and write a prose-format paper detailing everything a Cisco Networking Academy instructor should know about it. Incorporate configuration and verification commands.*

Example: What is Secondary IP Addressing?

Prompt: Act as a Cisco Routing expert. Explain the 5W1H of a Secondary IP Address and how to configure it. Create a detailed step-by-step lab with configuration examples. Make a one-paragraph lab overview, a topology using three Cisco 4331 routers, a complete IP addressing table, comprehensive instructions with examples, and finally, the complete raw configurations for all three routers.

Starting example:

"'

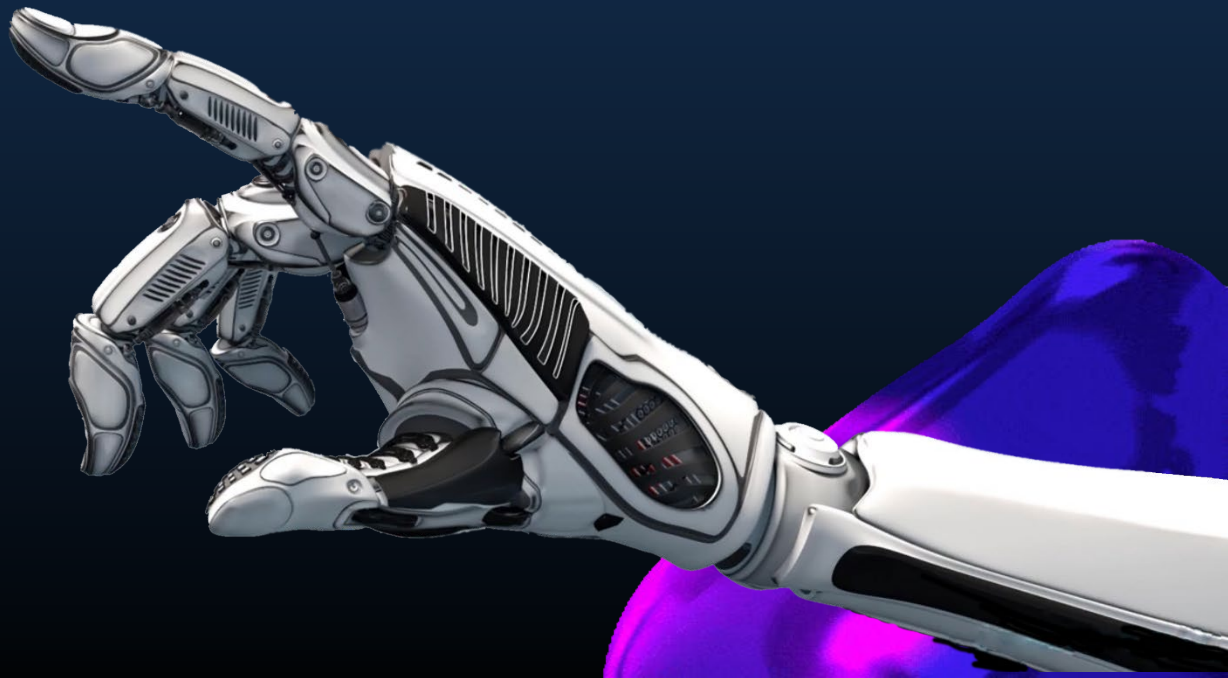
Add a secondary IP address to an interface:

```
Router(config)# interface gigabitEthernet0/0/0
 Router(config-if)# ip address 192.168.1.1 255.255.255.0
 Router(config-if)# ip address 10.10.10.1 255.255.255.0 secondary
 Router(config-if)# no shutdown
 Router(config-if)# exit
```

"'

[Completed Lab]

Question Prompting



Use AI to Create STP Practice Questions

Sample Prompt:

```
You are a CCNA instructor. Write a multiple-choice question about Spanning Tree root bridge
selection. Include switch priorities and MAC addresses. Provide the correct answer and a one-line
explanation. Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my
request. Then internally iterate on your work until it scores 10/10 against that rubric, and show
me only the final, perfect output.
```

Sample Response:

```
Question:
Four switches have the following priorities and MAC addresses:
Switch A: priority 32768, MAC 00:11:22:33:44:01
Switch B: priority 32768, MAC 00:11:22:33:44:02
Switch C: priority 4096, MAC 00:11:22:33:44:03
Switch D: priority 32768, MAC 00:11:22:33:44:04
Which switch will become the root bridge?
```

Correct Answer: Switch C

```
Explanation: STP selects the
switch with the lowest bridge ID
(priority + MAC); Switch C has the
lowest priority (4096).
```

Use AI to Generate NAT Concept Questions

Sample Prompt:

```
You are a CCNA instructor. Write a multiple-choice question about NAT behavior. Include inside
and outside IPs and provide the correct answer with an explanation. Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my
request. Then internally iterate on your work until it scores 10/10 against that rubric, and show
me only the final, perfect output.
```

Sample Response:

```
Question:
Which address is considered the inside global address in a NAT configuration?
A. 192.168.10.5
B. 203.0.113.5
C. 172.16.0.10
D. 10.0.0.1
```

Correct Answer: B

Explanation: Inside global is the public IP assigned to an internal device for communication over the Internet.

Challenge Student Reasoning with ACL Logic

Sample Prompt:

```
You are a CCNA instructor. Write a multiple-choice question that asks students to decide
which packet is denied by the ACL. Include port-based conditions. Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to
my request. Then internally iterate on your work until it scores 10/10 against that
rubric, and show me only the final, perfect output.
```

Sample Response:

```
ACL:
access-list 120 permit tcp any host 10.10.10.10 eq 22
access-list 120 permit udp any any eq 53
access-list 120 deny ip any any
Question:
Which of the following packets would be denied by this ACL?
A.  TCP from 192.168.1.5 to 10.10.10.10 on port 22
B.  UDP from 172.16.0.2 to 8.8.8.8 on port 53
C.  TCP from 192.168.0.1 to 10.10.10.10 on port 80
D.  UDP from 10.1.1.1 to 192.168.2.2 on port 53
```

Correct Answer: C

Explanation: Only SSH (TCP port 22) to host 10.10.10.10 and UDP DNS (port 53) are permitted; TCP port 80 traffic is denied.

Test Config Understanding with DHCP Questions

Sample Prompt:

```
You are a CCNA instructor. Write a multiple-choice question to test DHCP relay
configuration. Include a brief scenario and the correct answer. Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my
request. Then internally iterate on your work until it scores 10/10 against that rubric, and
show me only the final, perfect output.
```

Sample Response:

```
Question:
A router interface is configured as follows:
  interface G0/1
   ip address 192.168.100.1 255.255.255.0
   ip helper-address 10.10.10.10
What is the purpose of the ip helper-address command?
A.  It assigns a default gateway to the client
B.  It specifies the DHCP server to forward client requests to
C.  It assigns an IP to the router interface
D.  It configures the router to act as a DHCP server
```

Correct Answer: B

Explanation: The ip helper-address command forwards DHCP requests to a remote DHCP server.

Isomorphic Deterministic Questioning

Sample Prompt:

```
Using isomorphic deterministic questioning techniques, create 10 variations of this question, changing the IP address and/or subnet mask.
'''
What class is this IP address?
10.10.10.1 255.0.0.0
A.    Class A
B.    Class B
C.    Class C
D.    Class D
E.    Class E
'''
Show the correct answer after each question and a one-sentence explanation.
```

Sample Response:

```
What class is this IP address?
169.254.0.1 255.255.0.0
Class A
Class B
Class C
Class D
Class E
```

```
Answer: Class B
Explanation: The IP address 169.254.0.1 falls
within the Class B range (128.0.0.0 to
191.255.0.0), but it is specifically reserved
for Automatic Private IP Addressing (APIPA).
```

Isomorphic Deterministic Questioning

Sample Prompt:

```
Using isomorphic deterministic questioning techniques, create 5 variations of this question, changing the name, job title, the
scenario, and the routing protocol (RIP, EIGRP, OSPF) while keeping the main objective the same. Give 4 answer choices.
'''
Jim is a network administrator. He has been tasked with configuring EIGRP in a network consisting of 3 routers and 5 different
subnets. What command can he use to see if the network is configured correctly?
'''
Show the correct answer after each question and a one-sentence explanation. Think deeply about this.
Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on
your work until it scores 10/10 against that rubric, and show me only the final, perfect output.
```

Sample Response:

the routers are exchanging routing information?

•A) show ip protocols

•B) show ip eigrp neighbors

•C) traceroute 25.0.0.2

•D) show interfaces

Answer: B) show ip rip database

Answer: A) show ip eigrp topology

advertised, and other protocol-specific details.

information.

Isomorphic Deterministic Questioning

Create three prompts:

1. You pick the subject/objective and write the perfect question
2. Create 10 isomorphic Questions from the main question
3. Adjust your prompt until you get the questions you're looking for
4. Copy the completed prompt and questions into a document (.docx, .txt, .pdf)
5. Save to share at the end of class.
   Save as: *"Your Initials – Prompting Ideas"*

You have 10 minutes to complete.

Time Remaining:

Start

From Learning to Leading

How you can help students apply AI skills with confidence.

Ideas on Teaching Prompting in the Classroom

C.O.A.S.T Identification

- Give students good prompts to break apart.

Prompt Rewrites

- Give students vague prompts and ask them to improve clarity and structure.

Compare & Explain

- Show two outputs from different prompts. Have students identify which is better and why.

Prompt-Based Lab Tasks

- Replace a config task with a prompt challenge:
  - *"Write a prompt that generates the correct VLAN setup for this topology."*

Use AI as a Lab Assistant

- Have students check their configs by asking the AI:
  - *"What's missing from this?" or "Is this interface configured correctly?"*

Use AI to Simulate "show" Command Output

Sample Prompt:

```
You are a Cisco 4331 router. Simulate the output of show ip route on a device with:
GigabitEthernet0/0/0 in 192.168.1.0/24
GigabitEthernet0/0/1 in 10.0.0.0/8
OSPF-learned route to 172.16.0.0/16 via 10.1.1.1
Include route codes, administrative distance/metric, interfaces, and proper
formatting.
```

Sample Response:

```
Router# show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, O - OSPF, IA - OSPF inter area
* - candidate default, U - per-user static route

Gateway of last resort is not set

C        192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
C        10.0.0.0/8 is directly connected, GigabitEthernet0/0/1
O        172.16.0.0/16 [110/20] via 10.1.1.1, 00:00:12, GigabitEthernet0/0/1
```

Using Simulated Output for Teaching & Review

In-Class Use Ideas:

- Use simulated show output for concept review
  - *“What does this output tell you?”*
- Challenge students to spot missing or unusual entries
  - *“What’s one thing that’s not quite right in this output?”*

Turn it into a peer teaching activity

- One student writes a prompt, and another interprets the result

Teaching Tips:

- Use AI to simulate progressions. Start with a basic routing table, then add learned routes or static entries
- Have students prompt AI to generate output for their own topology diagrams

Show IP Route Questions

Use AI to Translate Configs into Plain Language

Sample Prompt:

```
You are a Cisco instructor. Explain this ACL configuration to a student new to networking:
'''
access-list 110 permit tcp 192.168.10.0 0.0.0.255 any eq 443
access-list 110 deny tcp 192.168.10.0 0.0.0.255 any eq 23
access-list 110 permit ip any any
'''
```

Sample Response:

This ACL allows devices in the 192.168.10.0/24 network to send TCP traffic to any destination if the destination port is 443, which is typically HTTPS.

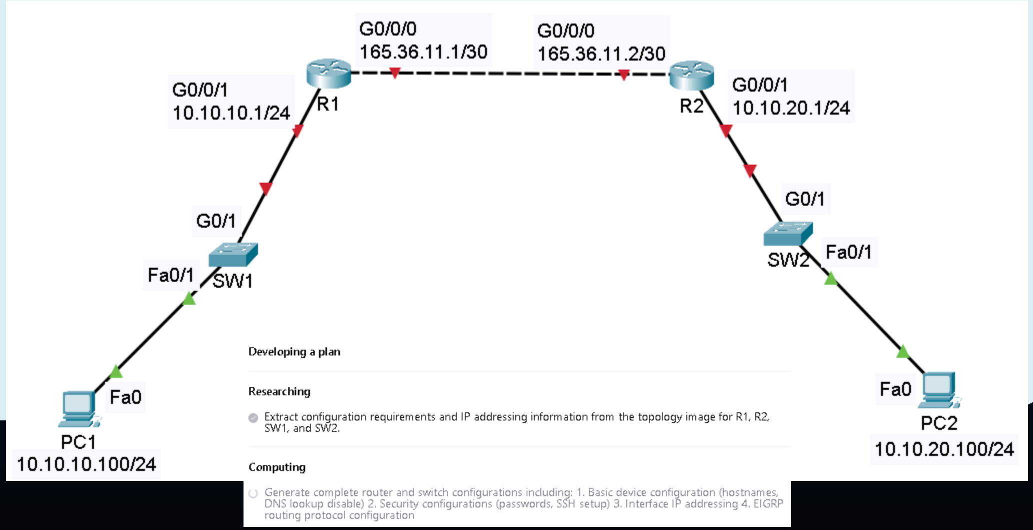
It then explicitly denies Telnet traffic (port 23) from that same source network to any destination. This is often done for security, since Telnet is unencrypted.

Finally, it permits all other IP traffic. This ensures that other non-blocked traffic is allowed through. In short, it allows secure web traffic, blocks Telnet, and permits everything else.

Use AI to Write a Configuration from a Graphic

Sample Prompt:

Act as a CCNA student. Create a complete raw configuration, with no comments or explanations, to configure hostnames, no lookups, secure all passwords, use SSH for local login to the console and vty ports, interface addressing, and use the EIGRP routing protocol to exchange all information.



Sample Response:

```
enable
configure terminal
hostname R1
no ip domain-lookup
ip domain-name ccna-lab.com
service password-encryption
enable secret Cisco123
username admin privilege 15 secret Admin123
crypto key generate rsa modulus 2048
ip ssh version 2
line console 0
 login local
 logging synchronous
 transport input ssh
line vty 0 4
 login local
 logging synchronous
 transport input ssh
 exit
interface GigabitEthernet0/0/0
 description Link to R2
 ip address 165.36.11.1 255.255.255.252
 no shutdown
interface GigabitEthernet0/0/1
 description LAN Network
 ip address 10.10.10.1 255.255.255.0
 no shutdown
 exit
router eigrp 1
 network 10.10.10.0 0.0.0.255
 network 165.36.11.0 0.0.0.3
 no auto-summary
end
copy running-config startup-config
```

Use AI to Generate IP Addressing Tables

Sample Prompt:

Act a a Cisco CCNA instructor. Generate three different IP addressing scenarios based on the provided IP addressing table. You can change the IP addressing scheme, ports, and number of routers as you see fit.

IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Description
R1	G0/0/0	25.0.0.1	255.255.255.252	Link to R2
R1	G0/0/1	192.168.1.64	255.255.255.224	Link to Local LAN
R1	Lo0	1.1.1.1	255.255.255.255	Link for Testing
R2	G0/0/0	25.0.0.2	255.255.255.252	Link to R1
R2	G0/0/1	30.0.0.1	255.255.255.252	Link to R3
R2	Lo0	2.2.2.2	255.255.255.255	Link for Testing
R3	G0/0/0	30.0.0.2	255.255.255.252	Link to R2
R3	G0/0/1	192.168.1.128	255.255.255.224	Link to Local LAN
R3	Lo0	3.3.3.3	255.255.255.255	Link for Testing

Sample Response:

Scenario 3: Add a fourth router (R4) to the topology and adjust the IP addressing scheme accordingly.

Device	Interface	IP Address	Subnet Mask	Description
R1	G0/0/0	192.168.0.1	255.255.255.252	Link to R2
R1	G0/0/1	10.0.0.1	255.255.255.0	Link to Local LAN
R1	Lo0	1.1.1.1	255.255.255.255	Loopback for Testing
R2	G0/0/0	192.168.0.2	255.255.255.252	Link to R1
R2	G0/0/1	192.168.1.1	255.255.255.252	Link to R3
R2	Lo0	2.2.2.2	255.255.255.255	Loopback for Testing
R3	G0/0/0	192.168.1.2	255.255.255.252	Link to R2
R3	G0/0/1	192.168.2.1	255.255.255.252	Link to R4
R3	Lo0	3.3.3.3	255.255.255.255	Loopback for Testing
R4	G0/0/0	192.168.2.2	255.255.255.252	Link to R3
R4	G0/0/1	10.0.1.1	255.255.255.0	Link to Local LAN
R4	Lo0	4.4.4.4	255.255.255.255	Loopback for Testing

Teaching with Explanation Prompts

In-Class Use Ideas:

- Give students a config and ask them to write:
  - *“Explain this as if you were teaching a new student.”*
    - Then compare their output to the AI’s
- Flip it: give them the AI’s plain-language response and ask:
  - *“What prompt would generate this?”*
- Use explanations to check for concept misunderstandings
  - *“What assumptions is the AI making? Are they valid?”*

Teaching Tips:

- Focus on building networking fluency, not just correctness
- Ask students to edit or improve the AI's explanation
- Use this format for peer reviews

[Teaching with Explanation Prompts]

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Teaching with AI-Generated Scenarios

In-Class Use Ideas:

- Use scenarios as review warm-ups or quiz questions
  - *“What’s the correct answer—and how do you know?”*
- Remove the explanation and have students justify their answer
  - Builds critical thinking and CCNA exam prep skills
- Have students write their own prompt to generate practice questions
  - Swap with peers to solve and discuss

Teaching Tips:

- Focus on reasoning, not just recall
- Let students compare AI’s logic to their own
- Use AI to remix scenarios by changing addresses, ports, or protocols

[Teaching with AI-Generated Scenarios]

Real-World Examples You Can Teach to Students

Prompts for Creativity and Critical Thinking Skills

"Act as a Cisco CCNA student preparing for the CCNA certification exam…"

1. **Real-World Networking Scenarios**: Ask students to create prompts that describe a real-world networking problem they might encounter, such as network congestion or security breaches.
   - *"Generate a list of troubleshooting steps for resolving network congestion in a corporate environment."*
2. **Networking Concepts Exploration**: Encourage students to write prompts that explore specific networking concepts.
   - *"Explain the differences between TCP and UDP in a way that a beginner can understand, including examples of when to use each."*
3. **Design a Network**: Have students create prompts that ask for network design suggestions based on specific requirements.
   - *"Design a small office network for a team of 10 people, including recommendations for hardware, IP addressing, and security measures."*

Prompts for Creativity and Critical Thinking Skills

4. **Security Best Practices**: Students can write prompts focused on cybersecurity.
   - *"List the top 10 best practices for securing a small business network against cyber threats."*
5. **Future of Networking**: Encourage students to think creatively about the future of networking.
   - *"Imagine a future where quantum computing is mainstream. Describe how this technology could change networking protocols and security measures."*
6. **Case Studies**: Students can create prompts based on hypothetical case studies.
   - *"Given a scenario where a company has experienced a data breach, outline a response plan that includes immediate actions and long-term strategies."*
7. **Networking Tools and Technologies**: Ask students to write prompts that explore various networking tools.
   - *"Compare and contrast three different network monitoring tools, highlighting their features, advantages, and disadvantages."*

Prompts for Creativity and Critical Thinking Skills

8. **Troubleshooting Scenarios**: Encourage students to write prompts that simulate troubleshooting scenarios.
   - *"You receive a call from a user who cannot connect to the Wi-Fi. Outline the steps you would take to diagnose and resolve the issue."*
9. **Networking Protocols**: Students can create prompts that delve into specific networking protocols.
   - *"Explain how the DHCP protocol works, including its role in IP address assignment and the process of lease renewal."*
10. **Networking Innovations**: Encourage students to think about innovations in networking.
    - *"Discuss the impact of Software-Defined Networking (SDN) on traditional networking practices and its potential benefits for businesses."*

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Prompts for Creating Labs

"Act as a Cisco CCNA student preparing for the CCNA certification exam…"

1. **Scenario-Based Prompts:** Encourage students to write prompts based on specific networking scenarios.
   - *"Create a lab that simulates a network outage due to a misconfigured router. Include detailed steps for troubleshooting and restoring connectivity."*
2. **Concept Exploration:** Have students write prompts that explore specific networking concepts or technologies. Students can research various networking concepts and create prompts that lead to lab designs that illustrate those concepts.
   - *"Design a lab that demonstrates the differences between static and dynamic routing protocols."*
3. **Real-World Applications:** Students can discuss current networking trends and write prompts that reflect real-world applications, encouraging practical learning.
   - *"Generate a lab that shows how to implement VLANs in a corporate environment to improve network segmentation."*

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Prompts for Creating Labs

4. **Troubleshooting Labs:** Students can identify common issues they've encountered and write prompts that guide the AI to create troubleshooting labs.
   - *"Create a lab that simulates a network performance issue and outlines the steps to diagnose and resolve it."*
5. **Security Focus:** Students can explore various security measures and write prompts that lead to labs focused on implementing those measures.
   - *"Design a lab that demonstrates how to configure firewall rules to protect a network from unauthorized access."*
6. **Collaborative Projects:** Students can work in teams to brainstorm and refine prompts, fostering collaboration and teamwork.
   - *"Develop a multi-site network lab that includes VPN configuration for secure remote access."*
7. **Simulation Tools:** Students can experiment with different simulation tools and write prompts that guide the AI to create labs tailored to those tools.
   - *"Generate a lab using Cisco Packet Tracer to simulate a basic network with multiple devices and configurations."*

Prompts for Creating Labs

8. **Performance Metrics:** Students can research performance metrics and write prompts that lead to labs designed to analyze those metrics.
   - *"Create a lab that measures the impact of different routing protocols on network latency and throughput."*
9. **Future Technologies:** Students can investigate future trends and write prompts that guide the AI to create labs that incorporate these technologies.
   - *"Design a lab that explores the implementation of Software-Defined Networking (SDN) in a virtualized environment."*
10. **Reflection and Feedback:** Students can create prompts that help them articulate their learning outcomes and provide feedback on the lab experience.
    - *"Generate a lab report template that includes sections for objectives, findings, and lessons learned from the lab."*

Prompts for Certification Study

"Act as a Cisco CCNA student preparing for the CCNA certification exam…"

1. **Create a Personalized Study Plan:** Ask students to write a prompt that generates a personalized study plan for the CCNA exam.
   - *"Create a 4-week study plan for the CCNA certification exam. Include daily topics, practical lab exercises, and review sessions. Focus on Network Fundamentals, IP Connectivity, and Security Fundamentals."*
2. **Generate Mock Exam Questions:** Have students write prompts to create mock exam questions for specific CCNA topics.
   - *"Generate 10 multiple-choice questions on VLAN configuration and troubleshooting. Include correct answers and explanations for each question."*

Prompts for Certification Study

3. **Simplify Complex Networking Concepts**: Encourage students to write prompts that simplify complex CCNA topics.
   - *"Explain the concept of Spanning Tree Protocol (STP) as if you were teaching it to a beginner. Use simple analogies and avoid technical jargon."*
4. **Design Lab Scenarios**: Ask students to write prompts that create hands-on lab scenarios for CCNA topics.
   - *"Create a lab exercise to configure and verify OSPF on three different Cisco routers in a small network. Include step-by-step instructions and expected outputs."*
5. **Troubleshooting Practice**: Have students write prompts to simulate troubleshooting scenarios.
   - *"Generate a troubleshooting scenario where a user cannot access the internet. Include possible causes related to IP addressing, DNS, and routing, and provide step-by-step solutions."*

Prompts for Certification Study

6. **Flashcard Creation**: Get students to write prompts that create flashcards for CCNA topics.
   - *"Create 20 flashcards for CCNA Network Fundamentals. Each card should have a question on one side and a detailed answer on the other."*
7. **Compare Networking Protocols**: Ask students to write prompts that compare different networking protocols.
   - *"Compare and contrast RIP, OSPF, and EIGRP routing protocols. Include their advantages, disadvantages, and use cases."*
8. **Subnetting**: Have students write prompts to generate subnetting quizzes.
   - *"Create a VLSM subnetting quiz with 5 questions. Include scenarios where I need to calculate subnet masks, usable ranges, network IDs, and broadcast addresses."*
9. **Explain Real-World Applications**: Encourage students to write prompts that connect CCNA topics to real-world scenarios.
   - *"Explain how VLANs are used in a corporate network to improve security and reduce broadcast traffic. Provide practical examples.*

AI Implementation

Implementation in Class:

- **Group Activities**: Have students work in groups to brainstorm and write prompts, then share them with the class for discussion.
- **Peer Review**: Have students share their prompts with peers for review and suggestions, fostering a collaborative learning environment.
- **Prompt Challenges**: Organize a challenge where students submit their best prompts, and the class votes on the most effective ones.
- **AI Interaction**: Use AI tools in class to generate responses to the prompts students create, allowing them to see the practical application of their writing.
- **Encourage Iteration**: Have students refine their prompts based on the quality of the AI-generated responses.

Classroom Ideas

Come up with five ways to incorporate AI into your classroom:

1. You pick the subject/objective
2. You pick the scenario.
3. Make at least one sample prompt for each scenario.
4. Copy the completed scenario and prompt into a document (.docx, .txt, .pdf)
5. Save to share at the end of class.
   Save as: *"Your Initials – Prompting Ideas“*

You have 15 minutes to complete.

Time Remaining:

Start

Creating Materials and Labs

Come up with a concept and/or lab you want to know about:

1. You pick the subject/objective
2. You pick the scenario.
3. Make a sample prompt and refine as needed.
4. Copy the completed prompt and scenario into a document (.docx, .txt, .pdf)
5. Save to share at the end of class.
   Save as: *"Your Initials – Prompting Ideas“*

You have 5 minutes to complete.

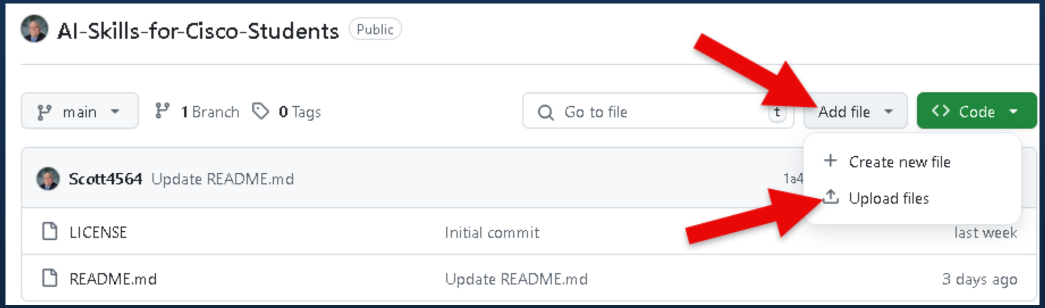
Time Remaining:

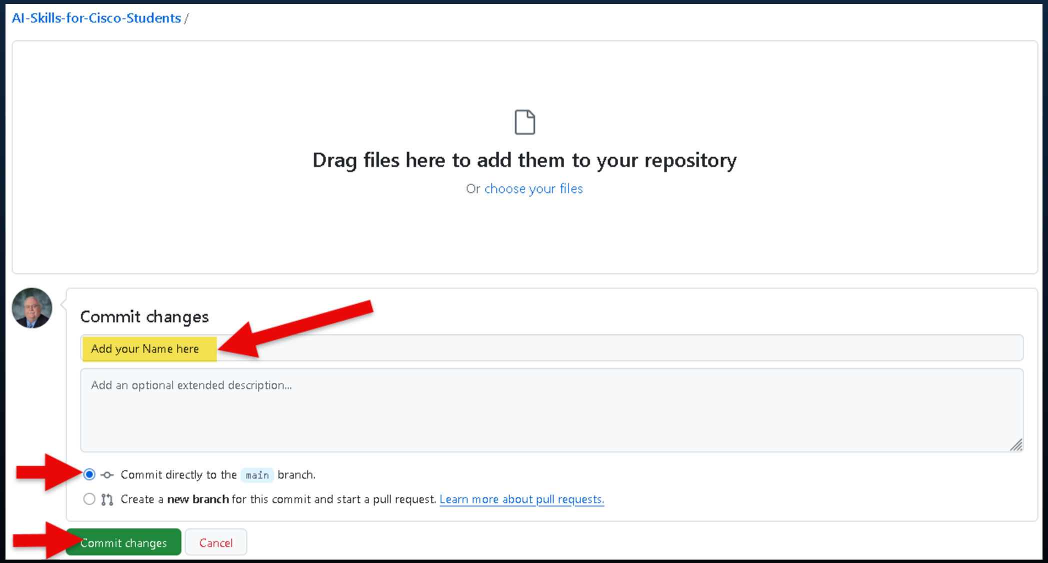
Start

Upload your Documents to GitHub

1. Open and log into GitHub
2. Navigate to the **`AI-Skills-for-Cisco-Students`** folder
3. Click on **`Add file -> Upload files`**
4. Drag files to browser
5. Add ***`Your Name`*** to Commit changes
6. Be sure `Commit` directly to the `main` branch is checked
7. Click **`Commit changes`**

`Note: You must have already added to the repository before you will have upload rights.`





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Where Do I Get My AI News?

- Superhuman AI - https://www.superhuman.ai/
- The Rundown AI - https://www.therundown.ai/
- Smarter with AI - https://www.smarterwithai.news/
- TLDR AI - https://tldr.tech/ai
- Decoding AI - https://www.decodingai.com

Want to Learn More?

Next Steps

DoL: AI Engineering Registered Internships

Overview

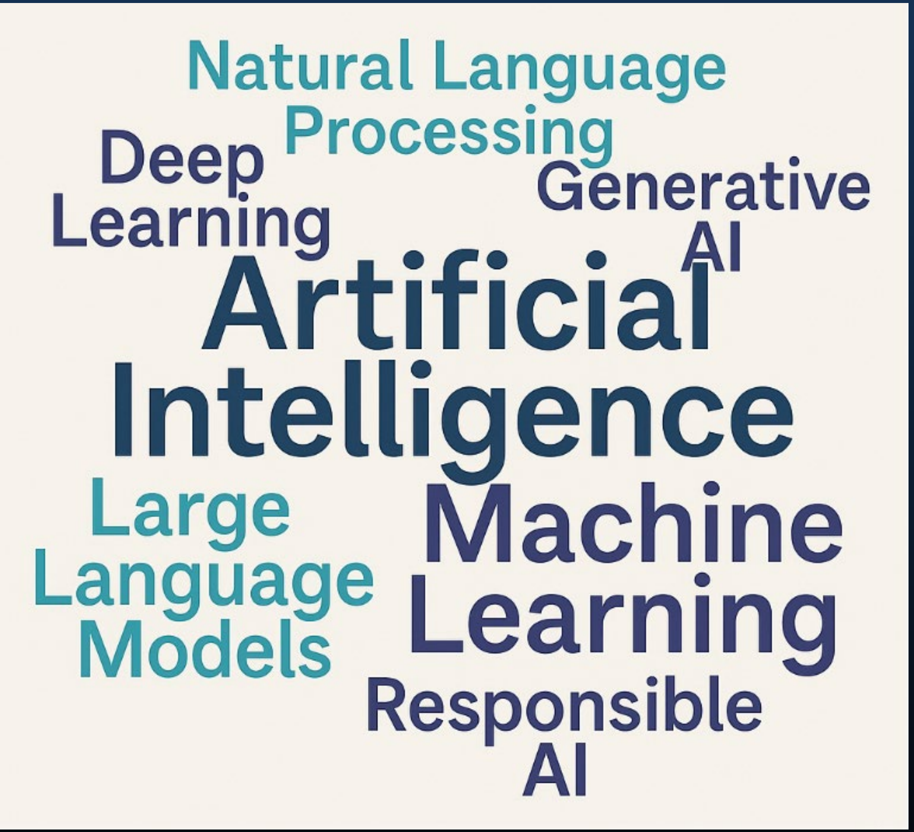
- Students will gain a comprehensive understanding of how AI and prompt engineering intersect with Cisco networking and automation, preparing them for current industry challenges and trends.
- To encourage active participation and collaboration to foster a rich learning environment, students will be expected to complete:
  - 4 to 6 hours per week for reading and labs
  - Plus 120-minute bi-weekly online Webex sync session
  - 36 weeks of study before starting an internship
  - 140 hours total

DoL: AI Engineering Registered Internships

AI for Network Engineers

7 Modules include:

1. **Foundations of AI** (4 weeks)
    - What is Artificial Intelligence?
    - Eras in Computing
    - A Brief History of Artificial Intelligence
    - What is Artificial Intelligence?
    - Key Concepts in AI
    - The Three Stages of AI Development
    - Data Types of AI
    - What is Machine Learning?
    - What is Deep Learning?
    - What is Natural Language Processing?
    - What is computer vision?
    - What is Generative AI?
    - What are Large Language Models?
    - Responsible AI
    - Structured Approach to Problem Solving
    - Current Trends in AI



DoL: AI Engineering Registered Internships

2. Linux, Cloud, Containers, and Version Control (2 weeks)

- What is Linux?
- What are Cloud Services?
- What is Docker?
- What is Version Control Using Git?
- What is an API?

DoL: AI Engineering Registered Internships

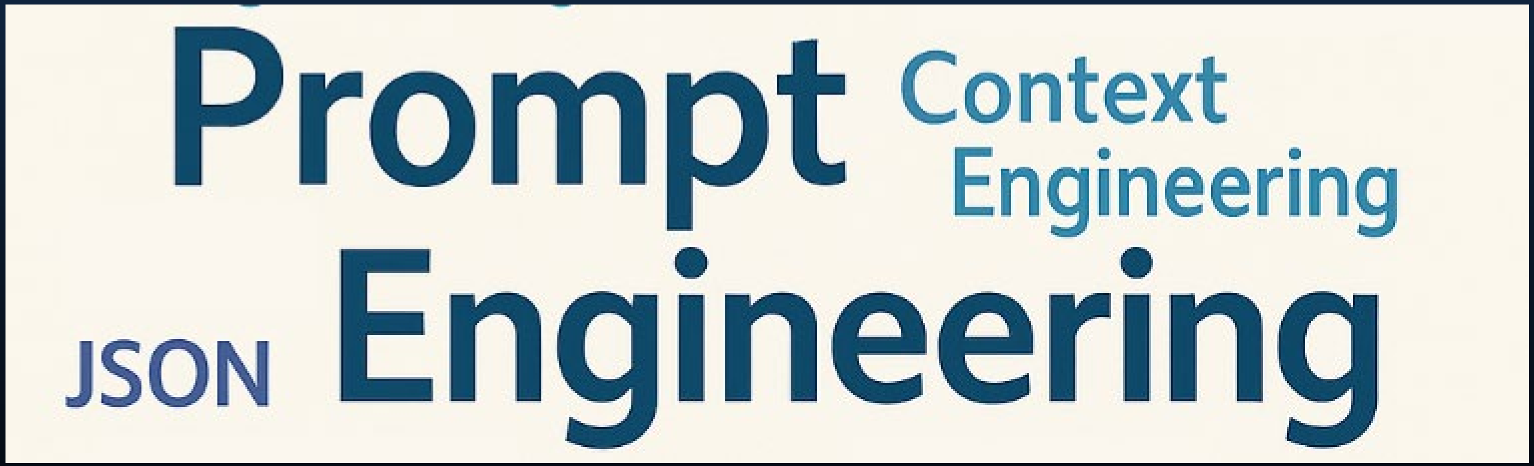
3. Python (6 weeks)

- Data Types, Variables, Operators, Formatting, and Basic I/O in Python
- Making Decisions in Python
  - Conditional Statements
  - Looping Constructs
- Data Structures in Python
  - Implementing Lists
  - Implementing Tuples
  - Implementing Sets
  - Implementing Dictionaries
- Functions in Python
- Scopes in Python
- Handling Text Files in Python
- Exceptions in Python
- Python Modules for AI

DoL: AI Engineering Registered Internships

4. Getting Started with Prompt Engineering (6 weeks)

- Introduction to Prompt Engineering
- Understanding Different Prompt Engineering Techniques
- Set up your machine for Prompt Engineering
- Prompt Engineering Formulas
- Prompt Engineering Techniques



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5. Prompt Engineering for Network Engineers (6 weeks)

- AI in Networking and Cybersecurity
- Automation with Python
- Automation with Ansible
- Automation with Terraform
- Automation with Nornir
- Basics of REST APIs
- Automating tasks using API calls
- Testing Configurations
- Automation Prompting

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6. AI Agents and Agentic AI (6 weeks)

- Introduction to AI Agents?
- AI Frameworks
  - LangChain
  - LangGraph
  - HuggingFace
  - N8N
  - LlamaIndex
  - Evaluation Metrics
- Introduction to RAG systems
- Introduction to MCP and A2A
- Introduction to Agentic AI
- Building AI Agents
- Building Agentic AI

DoL: AI Engineering Registered Internships

7. **Capstone Project** (6 weeks)
   - Create a capstone project that requires students to develop real-world applications to showcase their skills.
     - Develop an Agentic AI application that will automate configurations, monitor user interactions, troubleshoot, automatically fix issues, monitor and mitigate the network for security threats, and more.
     - Project Evaluation

Planned start date: January 2026

Thank you for leading the next generation of network engineers.

CISCO

Understand. Apply. Teach.

Elk Technologies

https://elktech.org

SCOTT STEPHENSON